

Knowledge, innovation, and technological development

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Economic Complexity, Class 3 – Oct 7 2025

Social Data Science, Corvinus University of Budapest

Economic complexity in Hausmann and Hidalgo 2009

Economic Complexity

The complexity (K_c) of an economy is the complexity (K_p) of its activities.
The complexity (K_p) of an activity is the complexity (K_c) of the economies where that activity is produced.



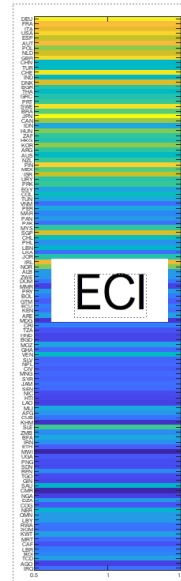
Complexity can be estimated as a solution to the following eigenproblem.

$$K_c = f(M_{cp}, K_p),$$

$$K_p = g(M_{cp}, K_c),$$

$$K_c = f(M_{cp}, g(M_{cp}, K_c)),$$

$$\tilde{M}_{cc'} K_{c'} = \lambda K_c$$



Hidalgo CA, Nature Review Physics (2021), PNAS (2009)

Definition	Working Name	Description: Short summary Question Form
$k_{a,0}$	Diversification	Number of products exported by country a . How many products are exported by country a ?
$\kappa_{\alpha,0}$	Ubiquity	Number of countries exporting product α . How many countries export product α ?
$k_{a,1}$	$k_{c,1}$	Average ubiquity of the products exported by country a . How common are the products exported by country a ?
$\kappa_{\alpha,1}$	$k_{p,1}$	Average diversification of the countries exporting product α . How diversified are the countries that export product α ?
$k_{a,2}$	$k_{c,2}$	Average diversification of countries with an export basket similar to country a . How diversified are countries exporting goods similar to those of country a ?
$\kappa_{\alpha,2}$	$k_{p,2}$	Average ubiquity of the products exported by countries that export product α . How ubiquitous are the products exported by product's α exporters?

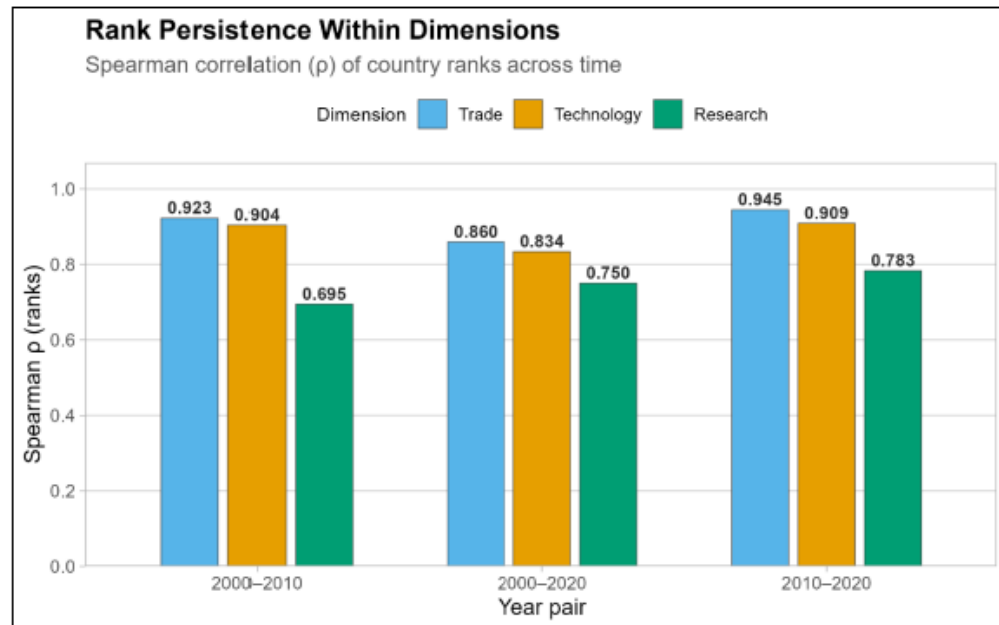
Table S 1 Interpretation of the bipartite network description obtained from the method of reflections.

Homework: correlations

	Trade_2000	Trade_2010	Trade_2020		Technology_2000	Technology_2010	Technology_2020		Research_2000	Research_2010	Research_2020
Trade_2000	1			Technology_2000	1			Research_2000	1		
Trade_2010	0.93	1		Technology_2010	0.83	1		Research_2010	0.84	1	
Trade_2020	0.86	0.95	1	Technology_2020	0.76	0.91	1	Research_2020	0.81	0.88	1

	Trade_2000	Technology_2000	Research_2000		Trade_2010	Technology_2010	Research_2010		Trade_2020	Technology_2020	Research_2020
Trade_2000	1			Trade_2010	1			Trade_2020	1		
Technology_2000	0.77	1		Technology_2010	0.74	1		Technology_2020	0.78	1	
Research_2000	0.49	0.43	1	Research_2010	0.70	0.69	1	Research_2020	0.55	0.56	1

Figure 2: Correlation matrices throughout time and dimensions



Source: Own work based on the given data

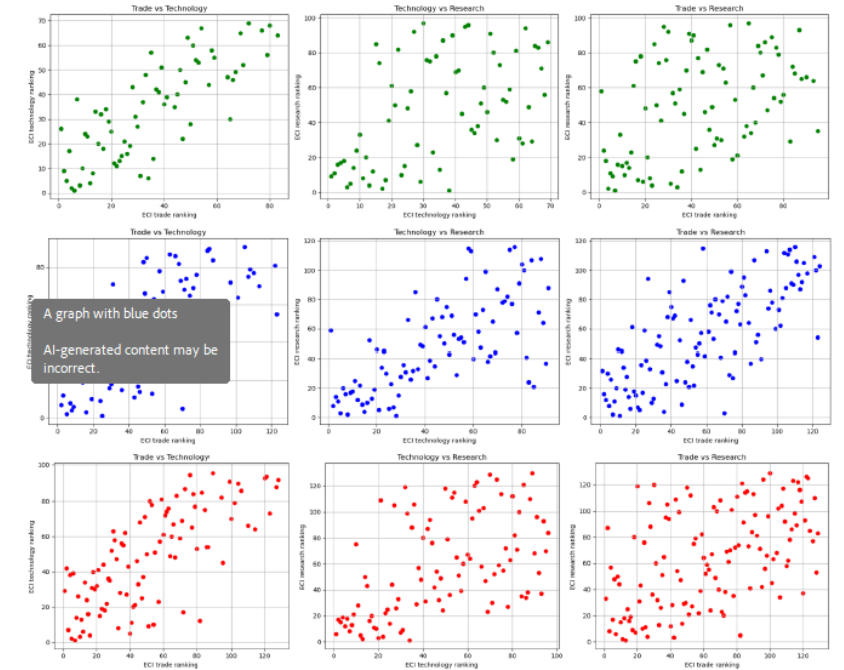


Figure 2: Figure 1: Comparison of ECI rankings between dimensions in 2000, 2010 and 2020

Homework: extras

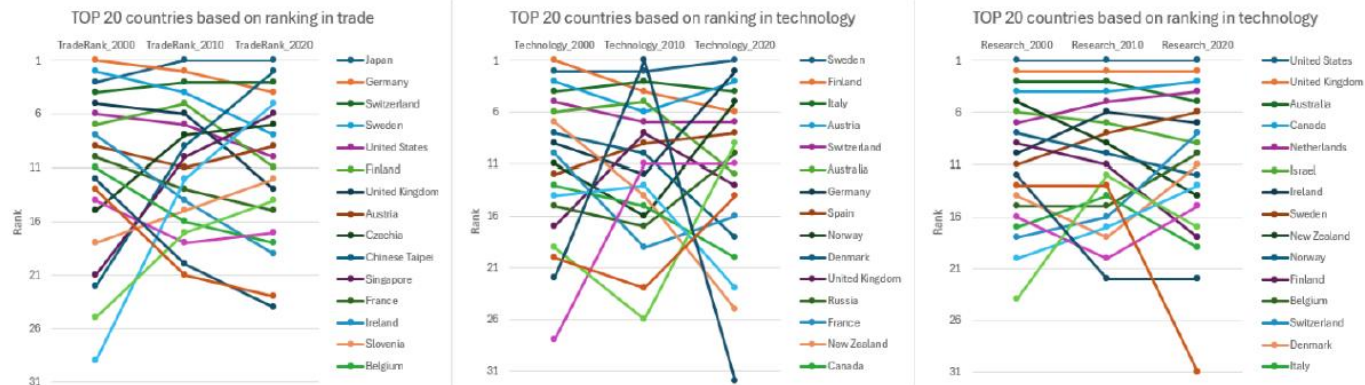


Figure 1: Top 20 countries based on the ranking of country complexity measure in 2000-2010-2020

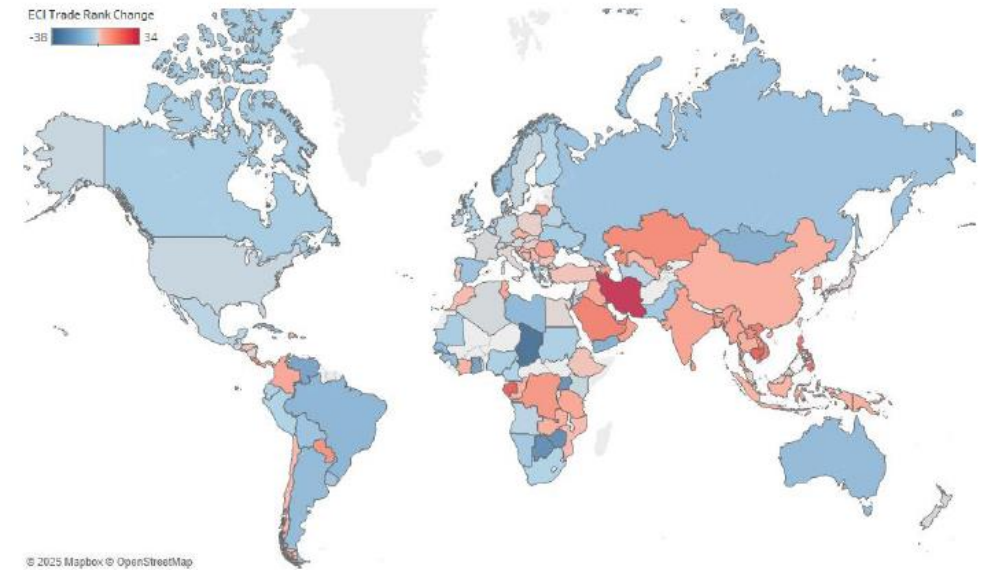
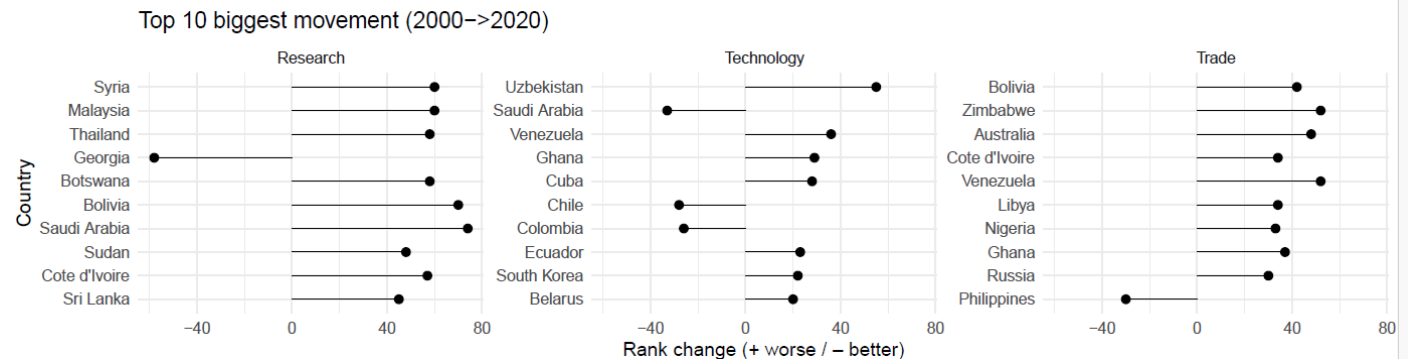


Figure 3: Changes in ECI Trade ranking from 2010 to 2020

Homework: extras

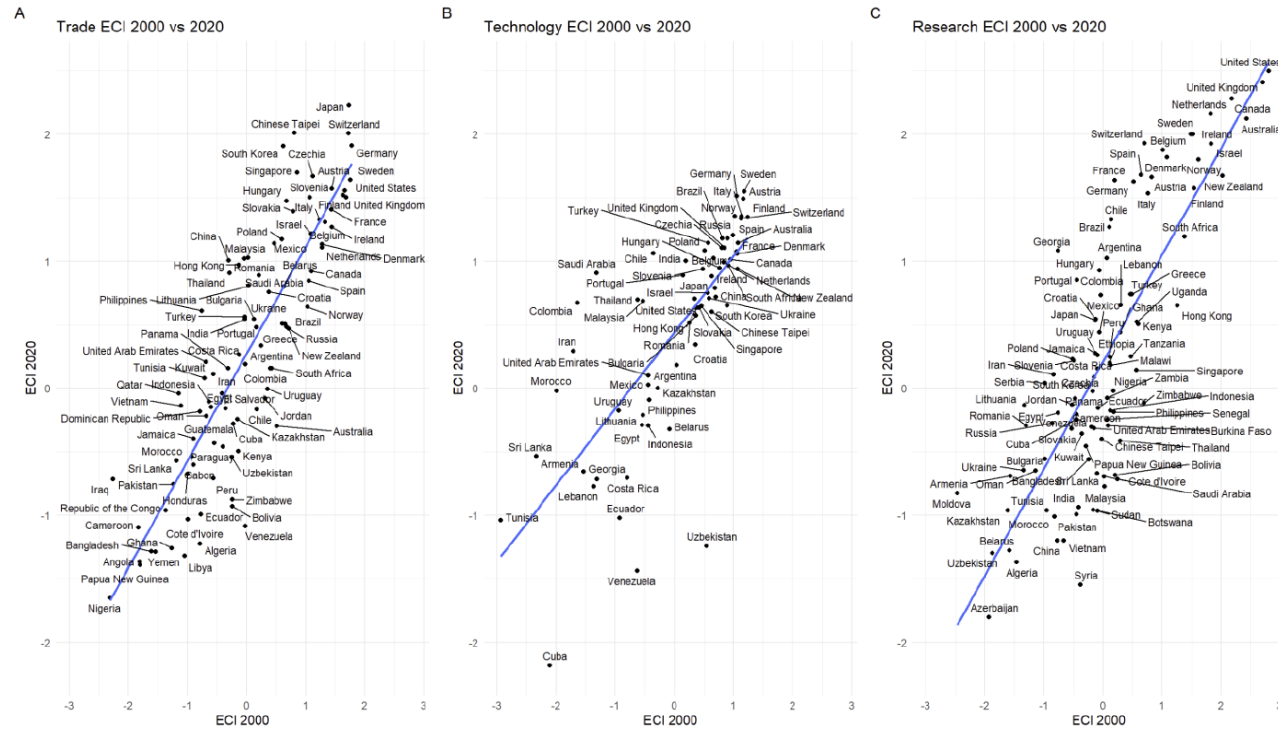
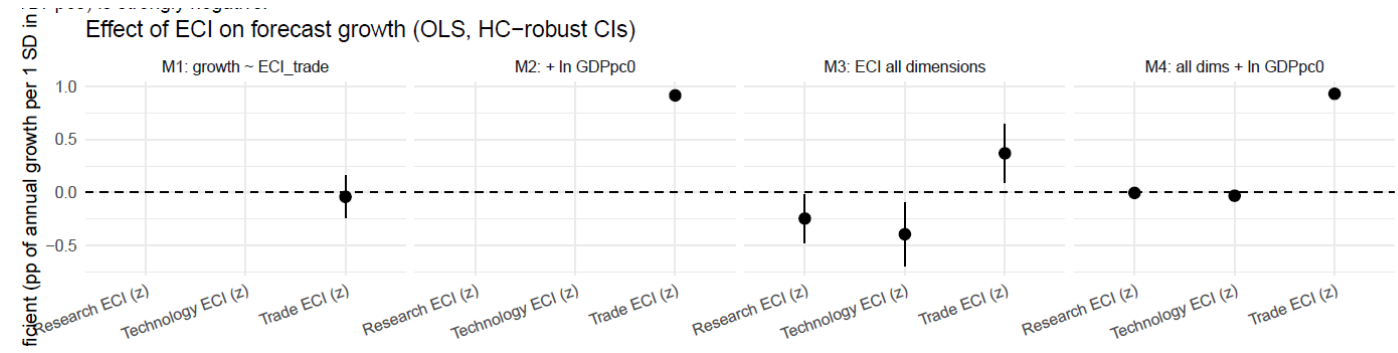
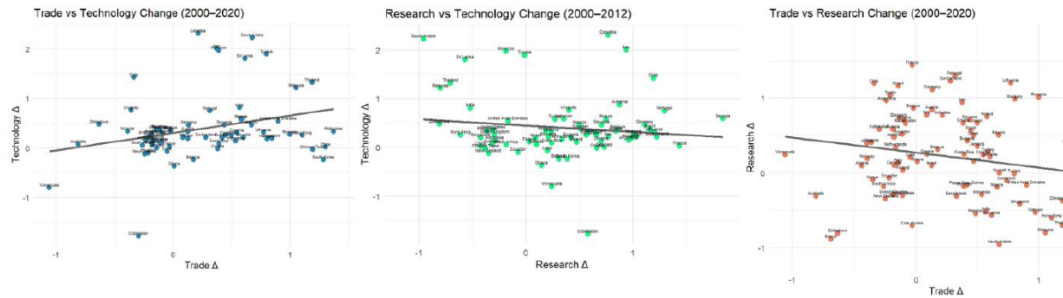


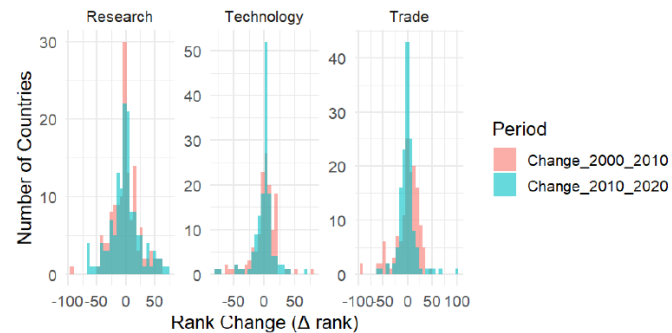
Figure 1: The relationship between the ECI in each category from 2000 to 2020

Homework: extras

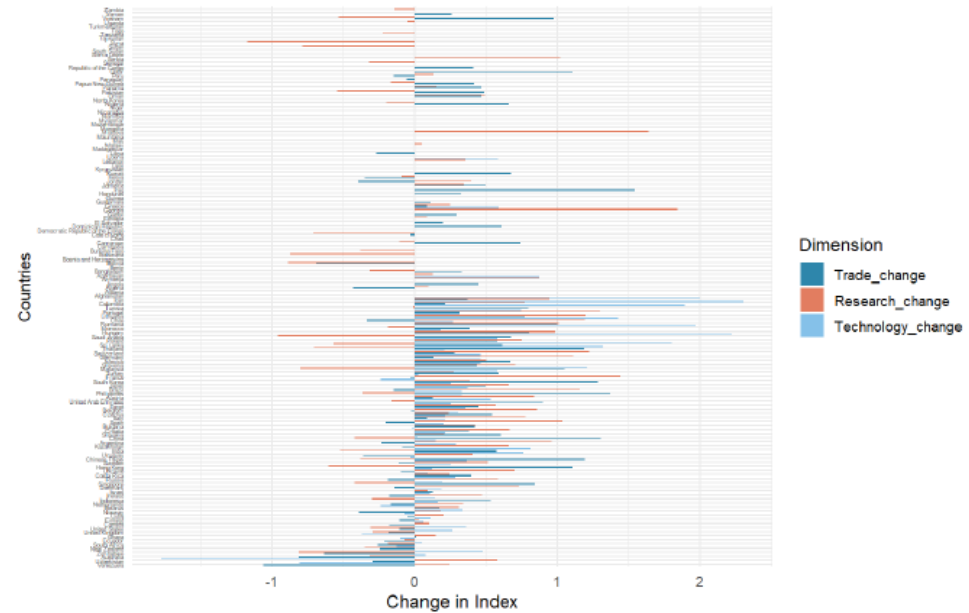


The three scatter plots illustrate how trade, technology, and research complexity changed over time. Between 2000 and 2020, there is a mild positive relationship between trade and technology growth, meaning that countries with more complex trade structures often also advanced technologically. However, the links between research and the other two dimensions are weakly negative, indicating that increases in research complexity did not consistently correspond with progress in trade or technology.

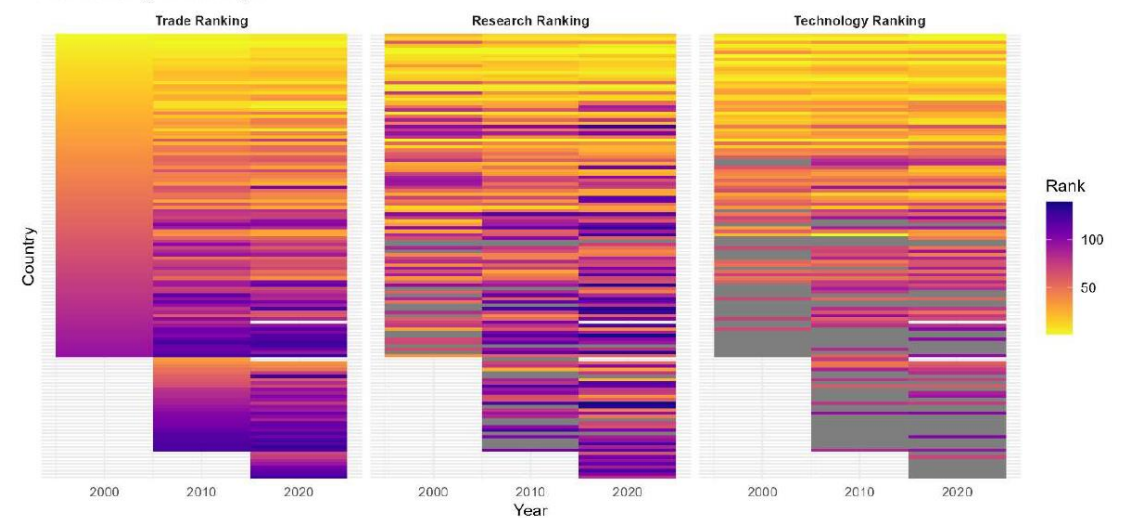
Distribution of Rank Changes (2000–2010–2020)



Change in Economic Complexity (2000–2020)



ECI Rankings Heatmaps



Homework: estimation

Regression Analysis

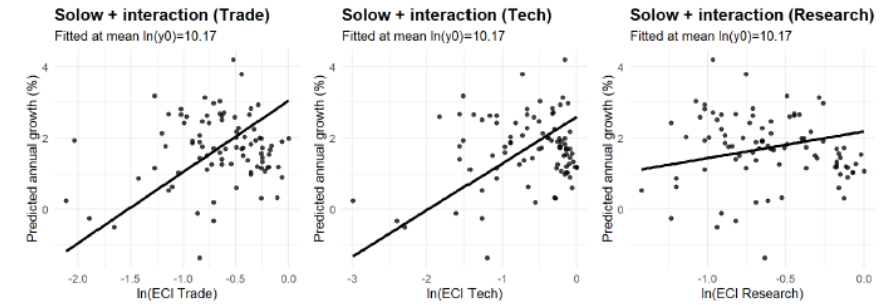


Figure 2

Using the Solow-style specification in Eq. (1), with predicted GDP per capita growth as the outcome and including $\ln(y_0)$, ECI, and their interaction, we find a consistent pattern across dimensions. The patterns can be seen in Figure 2.

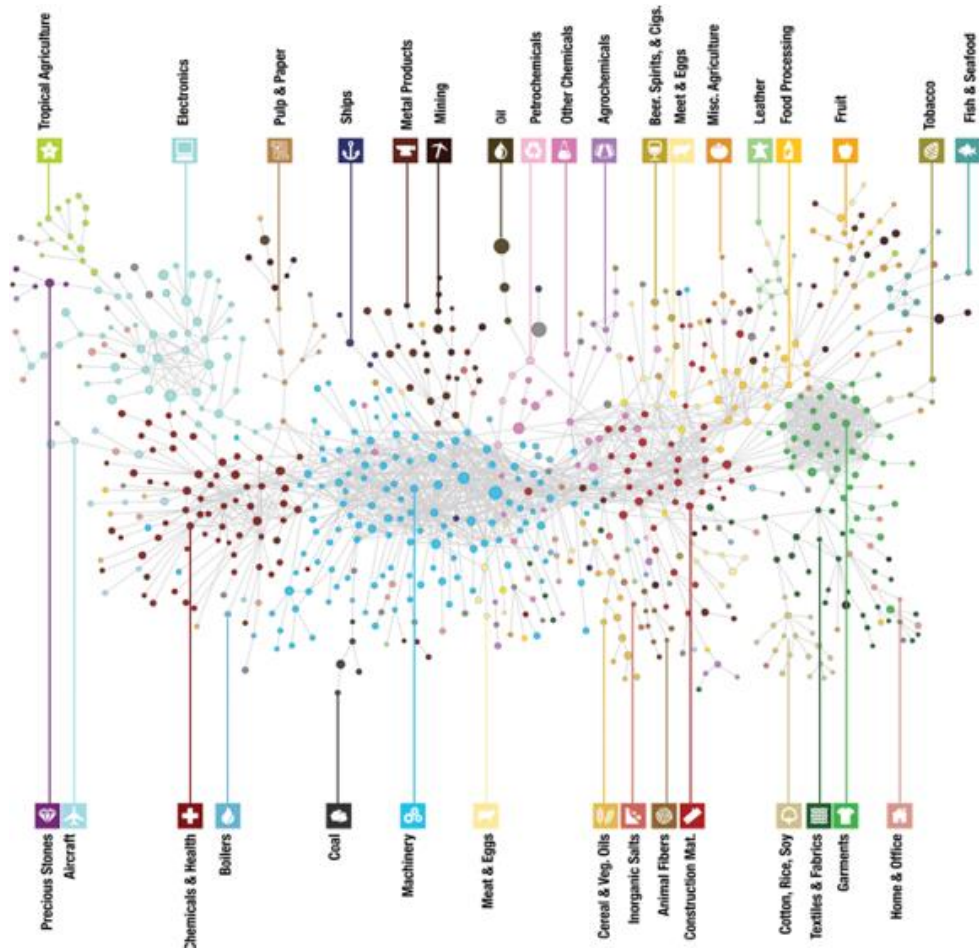
$$g_i = \beta_0 + \beta_1 \ln y_{0,i} + \beta_2 \ln(\text{ECI}_i) + \beta_3 \ln y_{0,i} \ln(\text{ECI}_i) + \varepsilon_i$$

Model estimates (Table 1) indicate that for trade and technology, the log-ECI main effect is positive and statistically significant at the 5% level of significance (Trade: $\hat{\beta}_2 \approx 4.63$, Technology: $\hat{\beta}_2 \approx 3.33$). In both models, the interaction term is negative and significant at the 10% level of significance (Trade: $\hat{\beta}_3 \approx -0.26$; Technology: $\hat{\beta}_3 \approx -0.20$), implying that the association between complexity and predicted growth attenuates as initial income rises. The convergence term is negative and highly significant in both cases ($\hat{\beta}_1 < 0$). By contrast, in the research model, neither the log-ECI main effect nor its interaction is statistically distinguishable from zero, and overall fit is markedly weaker (adjusted $R^2 \approx 0.35$). These patterns are consistent with the fitted lines: steep for Trade and Technology, comparatively flat for Research (though the slope does not matter here as it is statistically insignificant).

	Trade	Tech	Research
Intercept	18.86	16.85	12.38
$\ln(\text{ECI})$	4.62***	3.33***	4.04
$\ln(y_0)$	-1.56***	-1.40***	-1.00***
$\ln(y_0) * \ln(\text{ECI})$	-0.26*	-0.20*	-0.32
Adj. R^2	0.83	0.80	0.35
N	87	87	87

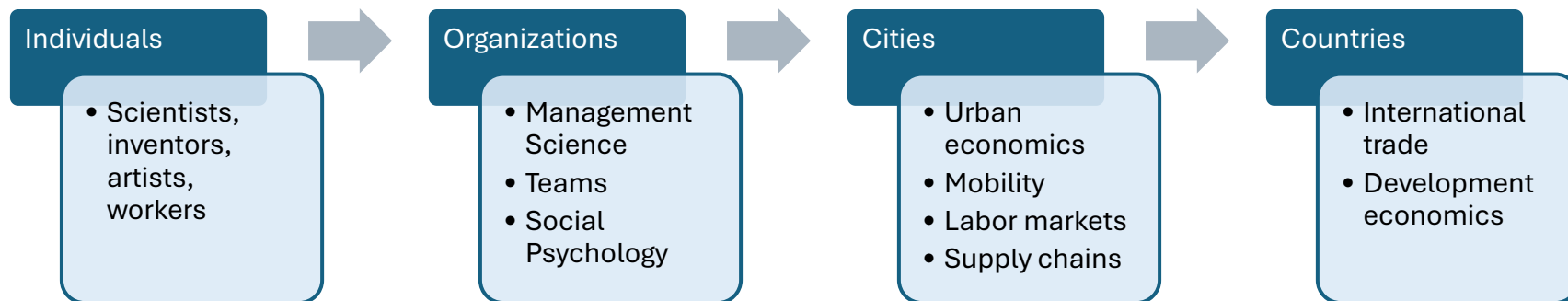
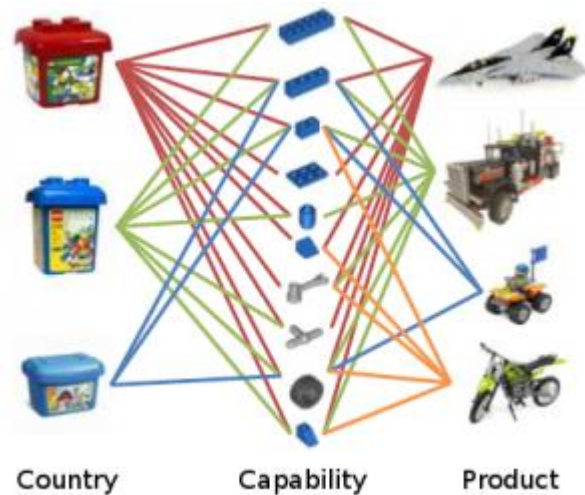
Table 1 * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

Product space: assumptions and messages



- Capabilities/knowledge
- Spatial specialization

Data-driven approach of economic complexity



The second block

3	Oct 7	Balázs Lengyel Knowledge, innovation, and technological development	II. Innovation and Economic Geography	[4]
4	Oct 14	Balázs Lengyel The development of cities and regions: agglomeration economies, the specialization-diversity debate		[5,6]
5	Oct 21	Zoltán Elekes Labor markets and skills: labor mobility, skill-relatedness		[7,8]
6	Nov 4	Balázs Lengyel Learning and innovation in networks: knowledge sourcing and dynamics of networks		[9,10]
7	Nov 11	Balázs Lengyel and Sándor Juhász Spatial and network inequalities: social capital, segregation and mixing		[11,12]
8	Nov 18	Presentation of students' research proposal		

[4] Dosi, G. (1988). Sources, procedures, and microeconomic effects of innovation. *Journal of Economic Literature*, 1120-1171.

[5] Audretsch, D. B., & Feldman, M. P. (1996). R&D spillovers and the geography of innovation and production. *The American Economic Review*, 86(3), 630-640.

[6] Glaeser, E. L., Kallal, H. D., Scheinkman, J. A., & Shleifer, A. (1992). Growth in cities. *Journal of Political Economy*, 100(6), 1126-1152.

[7] Neffke, F., Henning, M., & Boschma, R. (2011). How do regions diversify over time? Industry relatedness and the development of new growth paths in regions. *Economic Geography*, 87(3), 237-265.

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[10] Abbasiharofteh, M., Kogler, D. F., & Lengyel, B. (2023). Atypical combinations of technologies in regional co-inventor networks. *Research Policy*, 52(10), 104886.

[11] Tóth, G., Wachs, J., Di Clemente, R., Jakobi, Á., Ságvári, B., Kertész, J., & Lengyel, B. (2021). Inequality is rising where social network segregation interacts with urban topology. *Nature Communications*, 12(1), 1143.

[12] Aiello, L. M., Vybornova, A., Juhász, S., Szell, M., & Bokányi, E. (2025). Urban highways are barriers to social ties. *PNAS*, 122(10), e2408937122.

Today

1. Complex systems
2. Knowledge in economic thinking
3. Technological development
4. Measurement and recent research on innovation
5. Innovation in Central Europe

1. Complex systems

Complicated versus complex

Complicated



Complex



Complex systems

A **complex system** is a system made up of many interacting parts or components, where the collective behavior of the system is not easily predictable from the behavior of the individual parts.

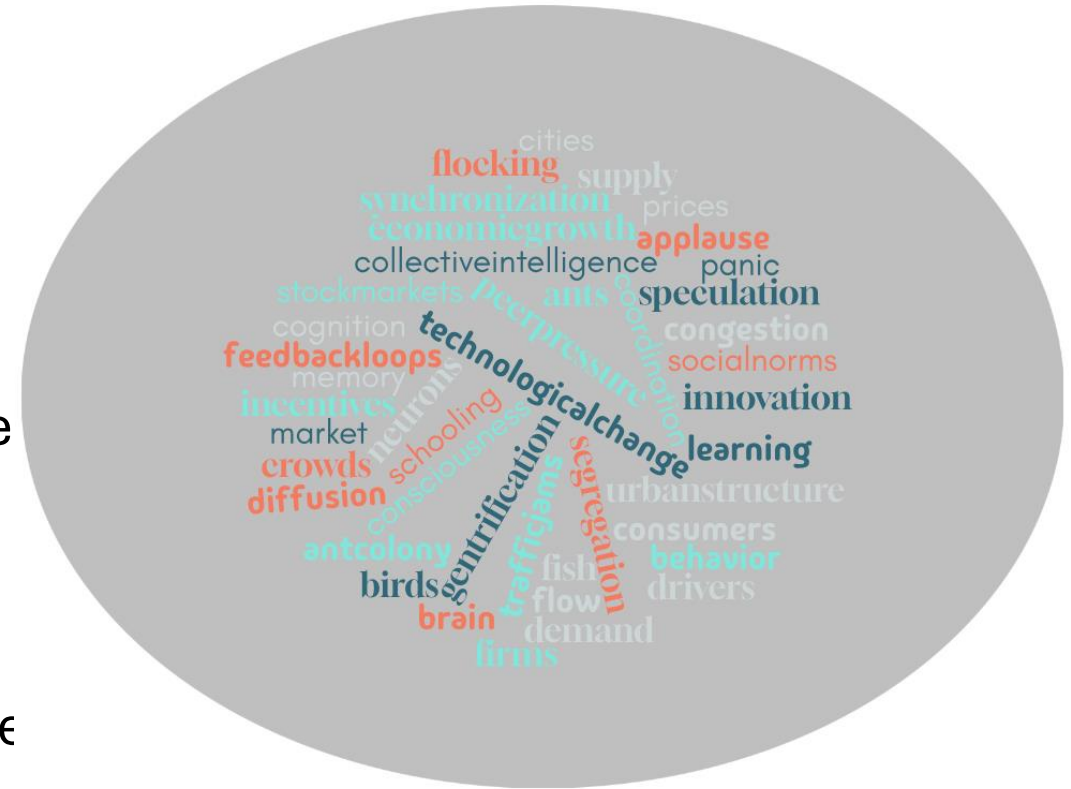
Decentralized decisions

Emergence

Patterns or behaviors emerge at the system level that are not present at the individual level.

Adaptation and feedback

The system evolves over time as components learn or adapt, often via feedback loops.



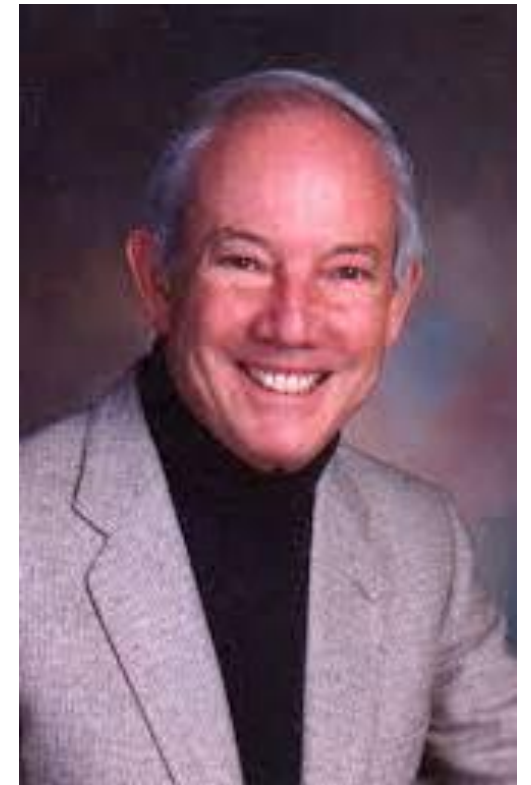
Murray Gell-Mann (SFI)
Complex adaptive systems
Nobel prize in physics 1969



Herbert Simon (Carnegie Mellon)
Bounded rationality
Nobel prize in economics 1978



Brian Arthur (SFI)
Path dependence



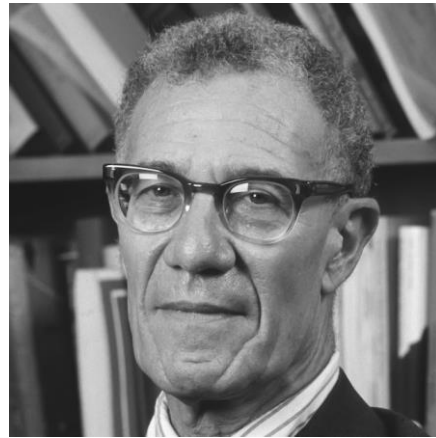
2. Knowledge in economic thinking

Technological development and growth

Robert Solow (MIT)

Technology in growth

Nobel prize in 1987



$$Y = F(K, AL) = K^\alpha (AL)^{1-\alpha}$$

$$A(t) = A(0)e^{gt}$$

g is exogenous technological development

Paul Romer (Princeton)

Endogenous growth

Nobel prize in 2018



$$L = L_A + L_Y$$

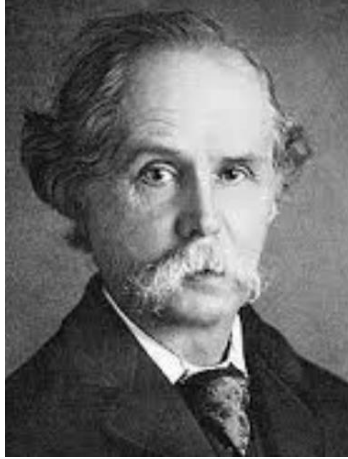
$$\dot{A} = \gamma L_A^\lambda A^\phi$$

$$Y = (AL_Y)^{1-\alpha} K^\alpha$$

Access to knowledge

Alfred Marshall
(Cambridge)

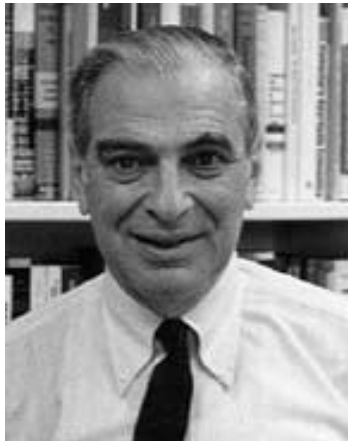
Knowledge is in the air



Kenneth Arrow (Harvard)

Learning by doing

Nobel prize in 1987



Matthew Jackson
(Stanford)

Learning in networks



Daron Acemoglu (MIT)

Nobel prize in 2024

Claude Shannon (MIT)

Information theory

A Mathematical Theory of Communication

By C. E. SHANNON

9. THE FUNDAMENTAL THEOREM FOR A NOISELESS CHANNEL

We will now justify our interpretation of H as the rate of generating information by proving that H determines the channel capacity required with most efficient coding.

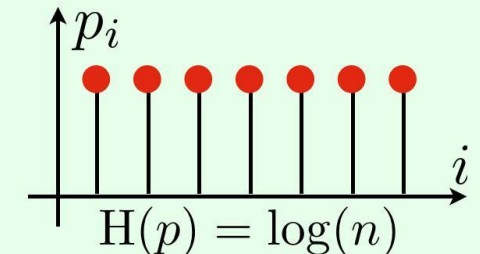
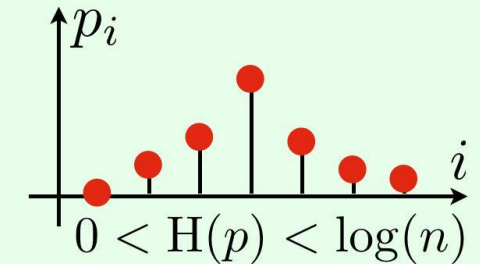
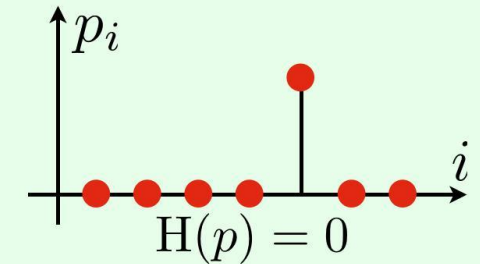
Theorem 9: Let a source have entropy H (bits per symbol) and a channel have a capacity C (bits per second). Then it is possible to encode the output of the source in such a way as to transmit at the average rate $\frac{C}{H} - \epsilon$ symbols per second over the channel where ϵ is arbitrarily small. It is not possible to transmit at an average rate greater than $\frac{C}{H}$.

Information theory is about information.
Information theory explains information.

Information theory quantifies uncertainty,
compresses data, and underlies modern
cryptography and machine learning.



$$H(p) \stackrel{\text{def.}}{=} - \sum_i p_i \log(p_i)$$



Michael Polanyi (Manchester)

Tacit knowledge



„We know more than we can tell.”

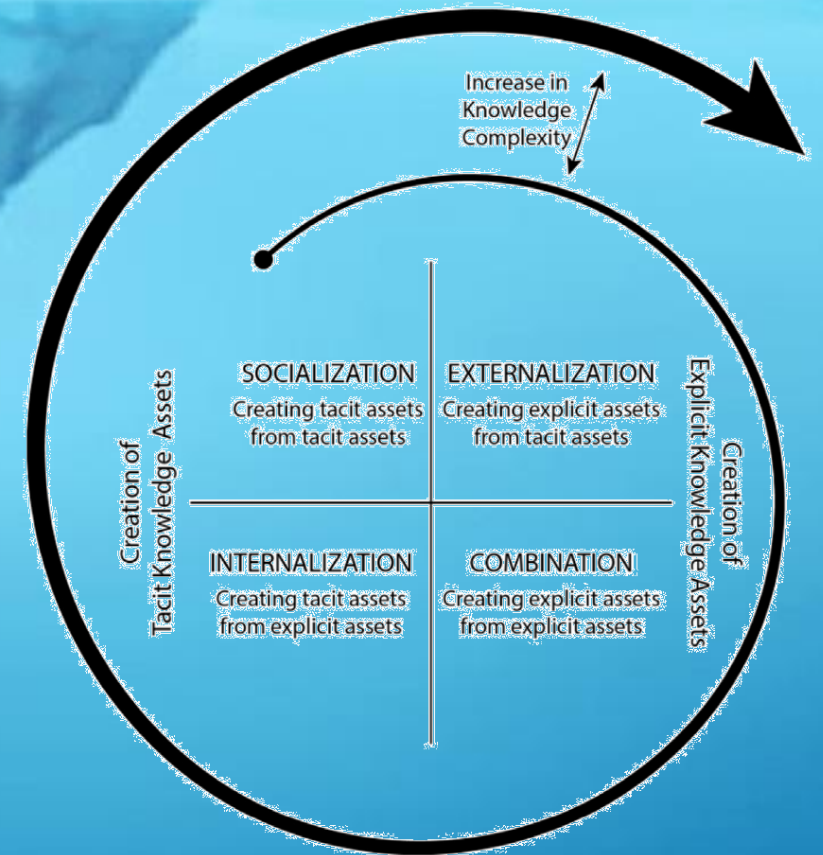
- **Personal and Experiential:** rooted in individual's experiences and shaped by specific context
- **Intuitive:** feelings, insights, hard to explain
- **Difficult to Document:** cannot easily be written down, spoken, or put into a standard procedure
- **Acquired Through Practice:** learned through doing, observing, and interacting
- **"Know-How" vs. "Know-That":** practical skill or understanding of how to do something, in contrast to explicit knowledge's focus on facts ("know-that")

Explicit Knowledge

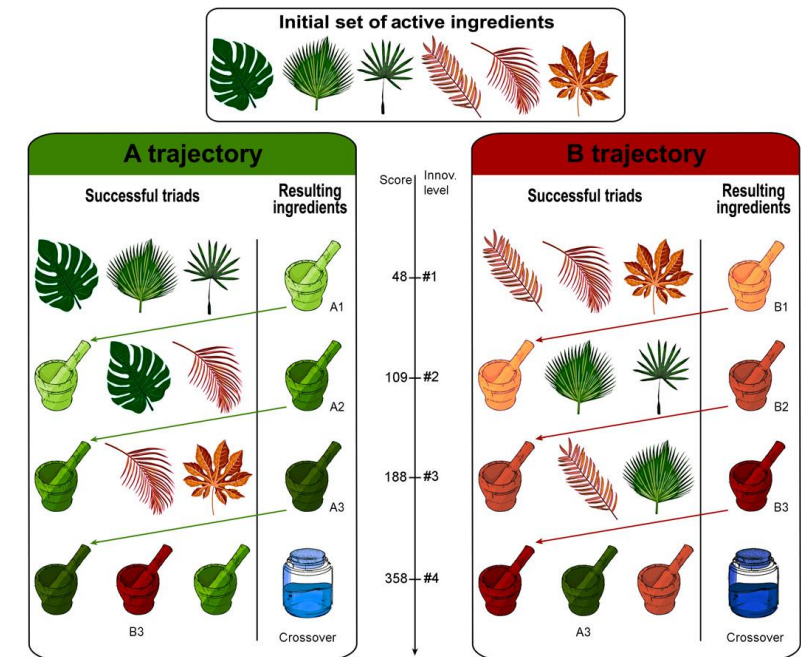
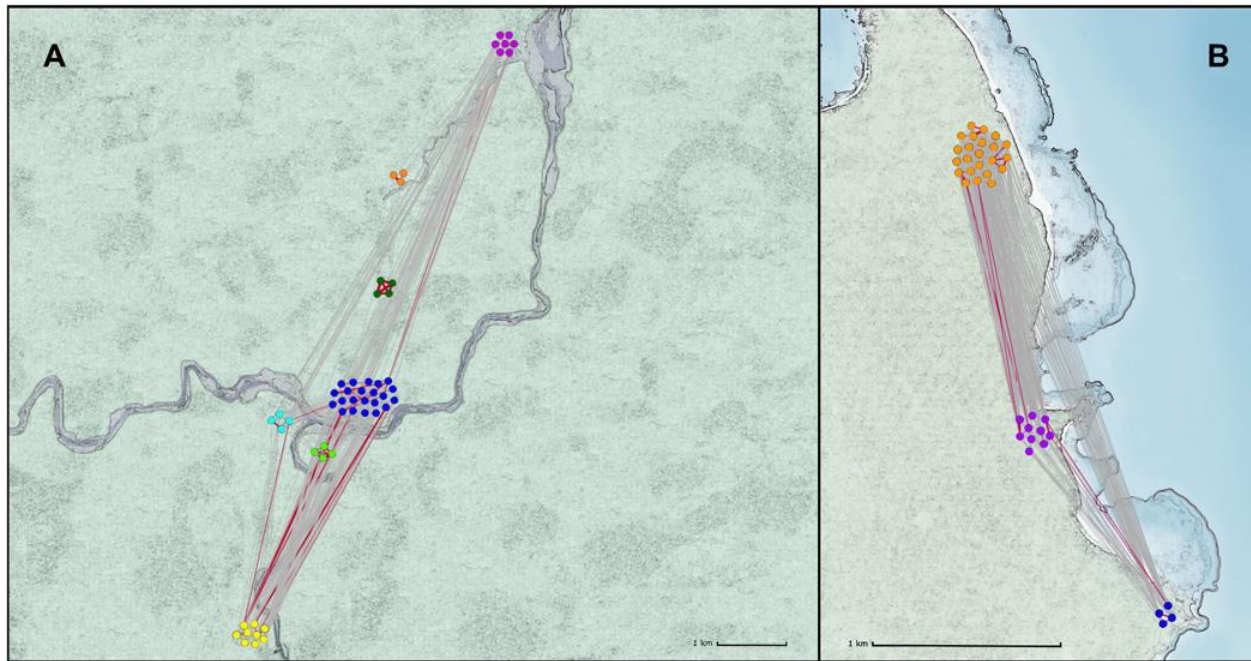
- Declarative information
- Rules & procedures

Tacit Knowledge

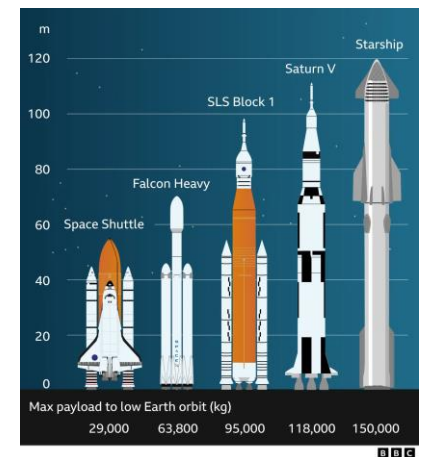
- Pattern recognition
- Perceptual discrimination
- Judging typicality
- Mental models
- Mindsets



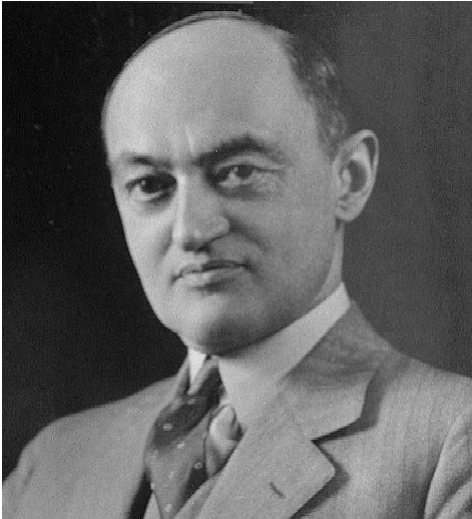
Learning-by-doing and mixing in networks



3. Technological development



Evolutionary economics



Joseph Schumpeter (Harvard)

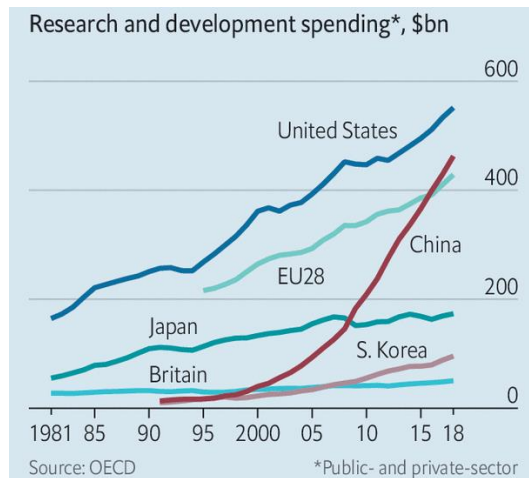
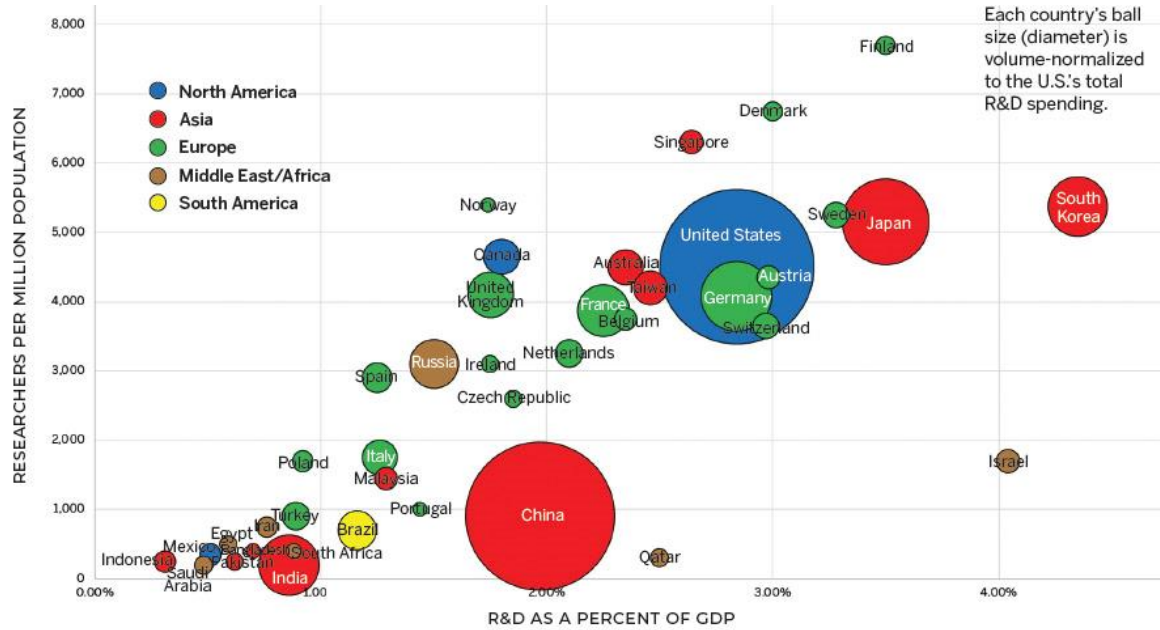
- Economic development is driven by innovation and technological development
- Creative destruction: old technologies are replaced by new one
- The Entrepreneur is the major driver of development

Concept in Biology	Schumpeterian Analogy
Mutation	Innovation
Natural selection	Market competition
Fitness	Profitability, survival
Extinction	Bankruptcy, firm failure
Speciation	Industry creation or transformation

Small versus Big

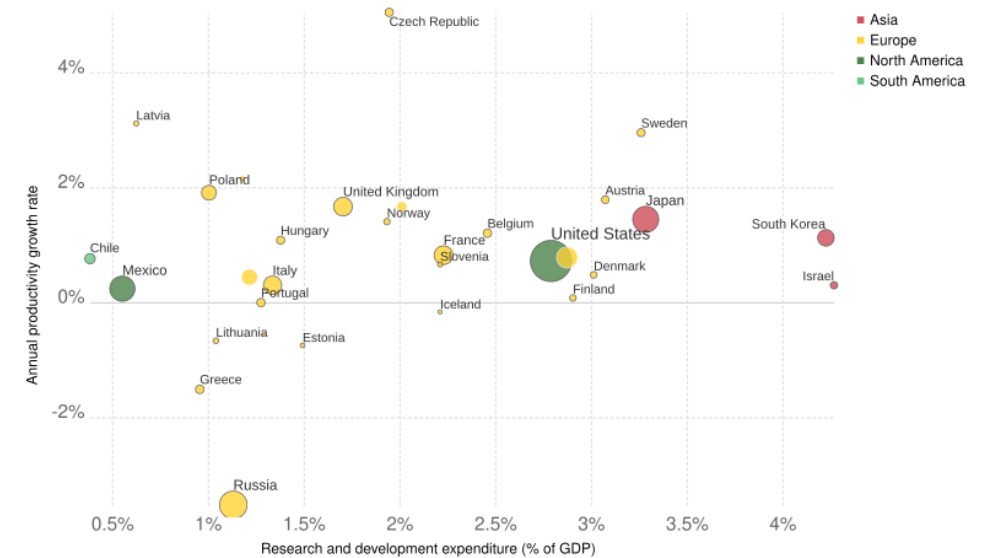


R&D facts



Annual productivity growth rate (%) vs. R&D expenditure (% of GDP), 2015
Annual growth in total economic productivity, measured as the annual change in gross domestic product (GDP) per

Annual growth in total economic productivity, measured as the annual change in gross domestic product (GDP) per hour worked (%) versus expenditure in research and development (R&D), measured as the share of total GDP.

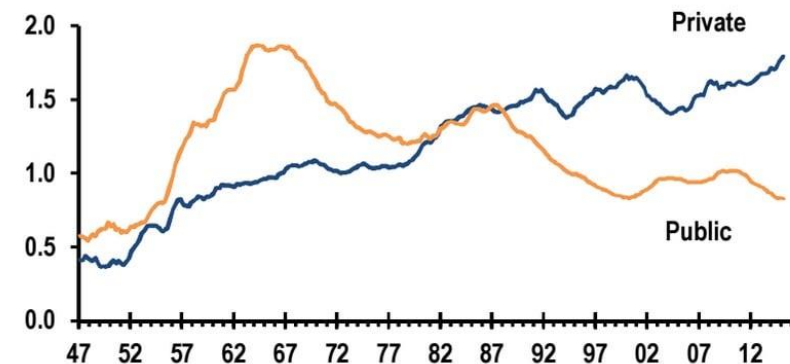


Source: Total Economy Productivity Growth - OECD, World Bank

CC BY

Figure 2: Public and private R&D spending

% of GDP

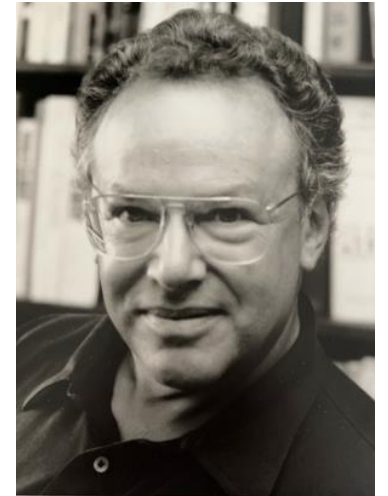


Source: BEA

Path dependence, lock-in

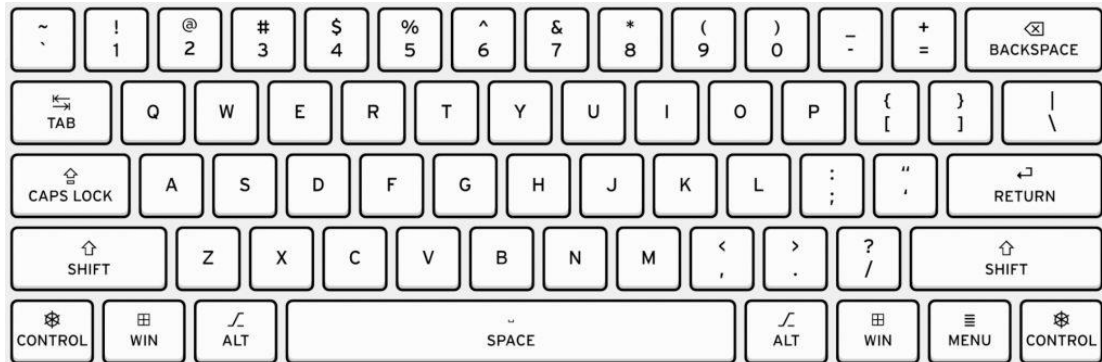


Paul David (Stanford)

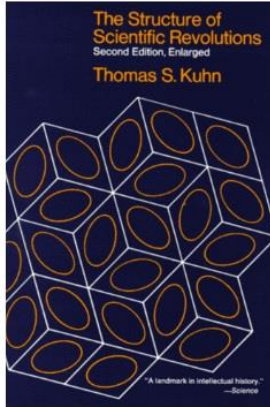
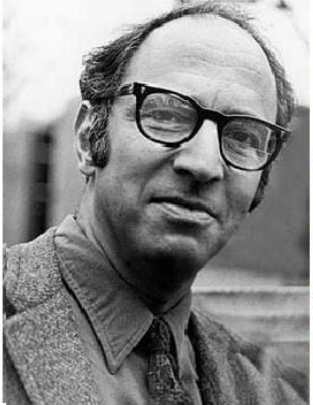


Key Mechanisms:

- **Historical accident:** QWERTY got early traction through deals with Remington, the dominant typewriter producer.
- **Learning effects:** People and institutions invested time and effort to learn QWERTY.
- **Network effects:** The more people used QWERTY, the more valuable it became (shared training, manuals, etc.)
- **Switching costs:** Moving to a new layout (like Dvorak) would require retraining, software changes, and user resistance — even if it's better.
- Result: **QWERTY is "locked in"** — not because it's the best, but because the **costs of switching** and the **advantages of compatibility** outweigh the benefits of alternatives.



Paradigms



Thomas Kuhn (Berkeley)

Scientific paradigms

- Most improvements are incremental
- Scientific breakthrough: evidence that our previous view is false

Giovanni Dosi (SPRU and Pisa)

Technological paradigms



Mechanism

Heuristics

Trajectories

Lock-in

Institutions

Selective attention

Shifts

Role in Paradigm

Guides incremental innovation

Channel long-term development

Reinforces dominance of the paradigm

Provide training, rules, and funding

Filters which problems are “worth” solving

Driven by crises, new knowledge, or unmet needs

The History of INNOVATION CYCLES

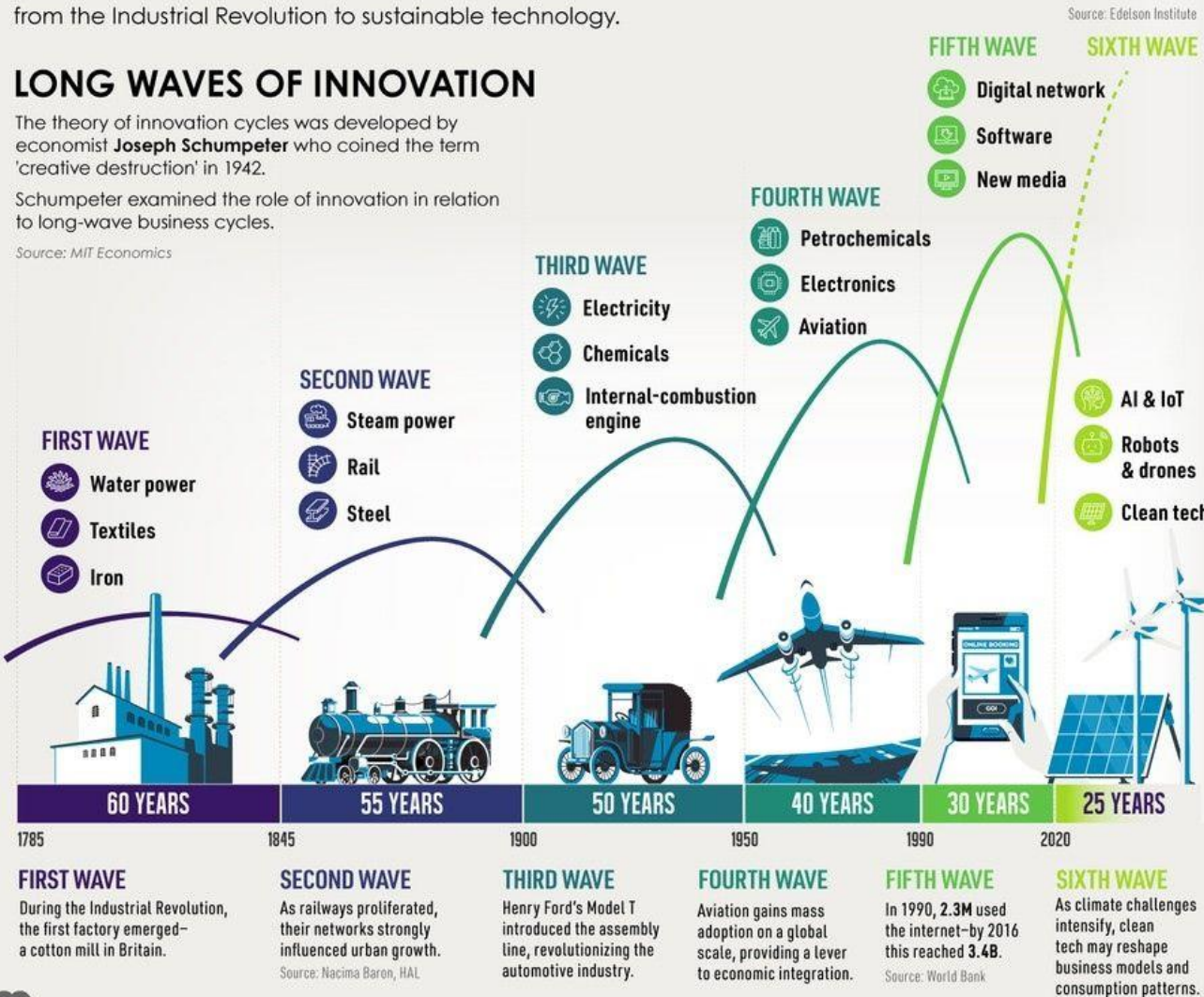
Below, we show waves of innovation across 250 years, from the Industrial Revolution to sustainable technology.

LONG WAVES OF INNOVATION

The theory of innovation cycles was developed by economist **Joseph Schumpeter** who coined the term 'creative destruction' in 1942.

Schumpeter examined the role of innovation in relation to long-wave business cycles.

Source: MIT Economics



Cycles of general purpose technologies (Perez 2002)

Mechanism

Technological paradigm

Financial speculation

Overinvestment

Crisis and correction

Institutional adaptation

Saturation and stagnation

Social pressure

Role

Sets the direction for innovation

Accelerates diffusion, but causes bubbles

Builds infrastructure, though inefficiently

Aligns tech and society post-bubble

Enables full deployment of GPT

Signals the end of a cycle

Drives the shift to the next paradigm

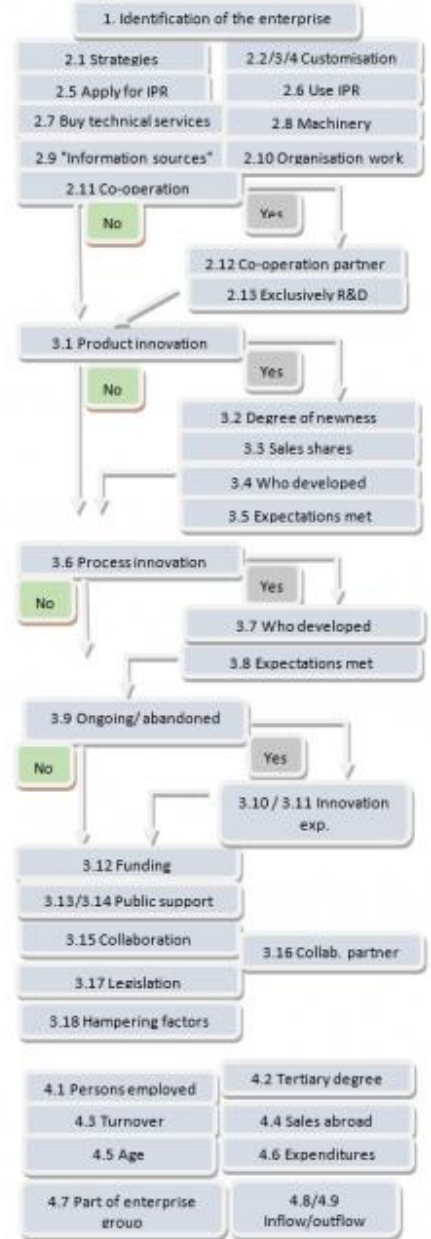
4. Measurement and recent research on innovation

Measurements of surveys

Flow of questions in CIS 2016



Flow of questions in CIS 2018



Measurement by statistical offices



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 English

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 Information

Eurostat is gradually releasing its metadata with a new design and layout. During this transition time, there will be a mix between metadata in the old and new look. To find out more, read our [news article](#).



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SCIENCE, TECHNOLOGY, AND INNOVATION

Database

Overview

Database

Visualisations

Publications

Information on data

Methodology

Quality

Legislation

Selected datasets

Science and technology (t_scitech)

Research and development (R&D) (t_rd)

High-tech industry and knowledge-intensive services (t_htec)

Human Resources in Science & Technology (t_hrst)

Detailed datasets

Science and technology (scitech)

Research and development (R&D) (rd) [\(Information note\)](#)

Community innovation survey (inn) [\(Information note\)](#)

High-tech industry and knowledge-intensive services (htec)

Human Resources in Science & Technology (hrst)

Intellectual property rights - Patents (pat)

Measurement from documents

(19)

(11)**EP 3 155 128 B1**

EUROPEAN PATENT SPECIFICATION

(45) Date of publication and mention of the grant of the patent:
15.05.2019 Bulletin 2019/20

(21) Application number: **15732984.8**

(22) Date of filing: **10.06.2015**

(51) Int CL:
C12Q 1/68 (2018.01)
C12N 15/11 (2006.01)
A61K 48/00 (2006.01)

(86) International application number:
PCT/NL2015/050420

(87) International publication number:
WO 2015/190921 (17.12.2015 Gazette 2015/50)

(54) **METHODS FOR CHARACTERIZING ALTERNATIVELY OR ABERRANTLY SPLICED MRNA ISOFORMS**
VERFAHREN ZUR CHARAKTERISIERUNG ALTERNATIV ODER ANOMAL GESPLEISSTER MRNA-ISOFORMEN
PROCÉDÉS DE CARACTÉRISATION D'ISOFORMES D'ARNM ÉPISSÉS DE MANIÈRE DIFFÉRENTE OU ABERRANTE

(84) Designated Contracting States:
AL AT BE BG CH CY CZ DE DK EE ES FI FR GB GR HR HU IE IS IT LI LT LU LV MC MK MT NL NO PL PT RO RS SE SI SK SM TR

(30) Priority: **10.06.2014 PCT/NL2014/050375**
04.09.2014 EP 14183623

(43) Date of publication of application:
19.04.2017 Bulletin 2017/16

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EP 3 155 128 B1

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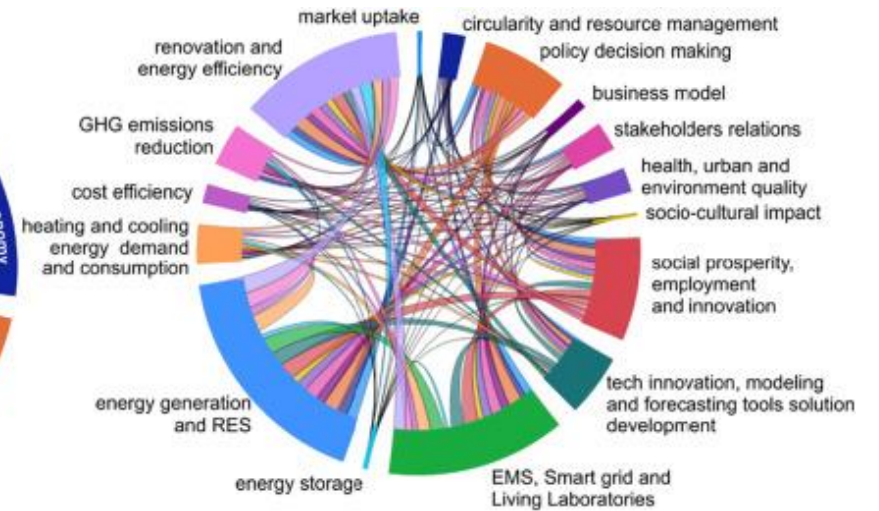
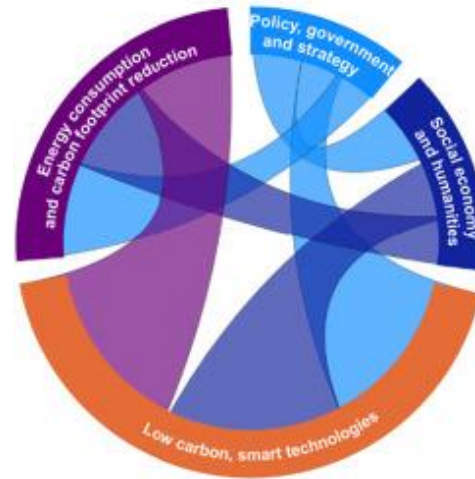
Inventors

Representative

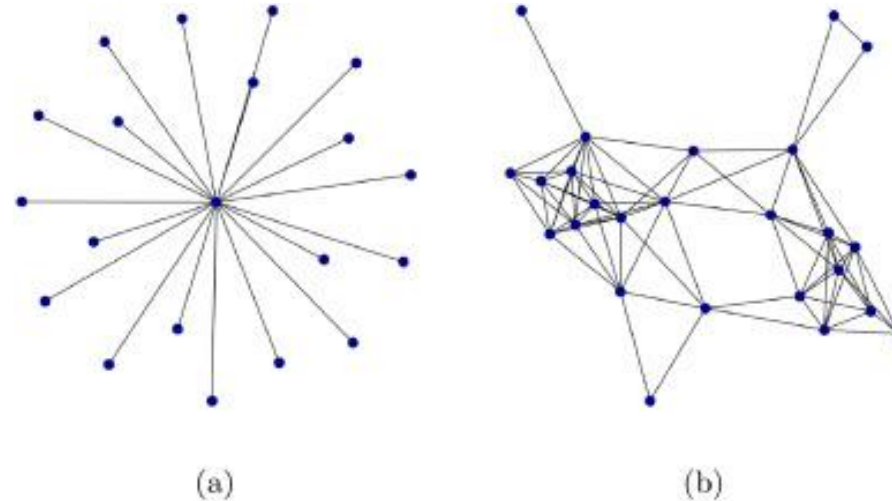
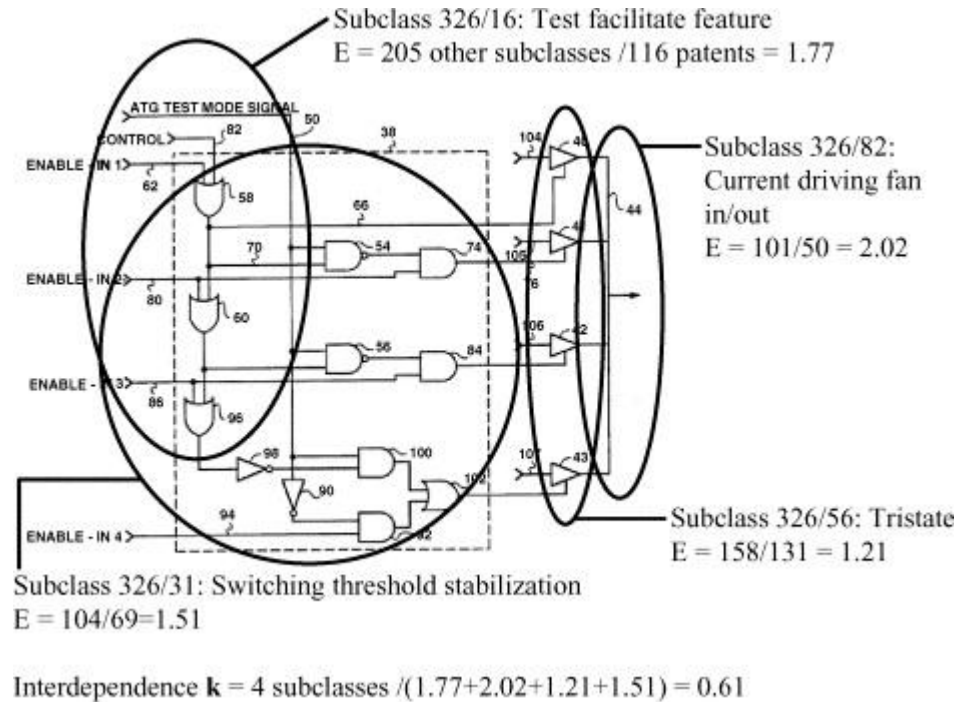
Publication number

Patent classes

Research grants



Combination of knowledge as measurement of knowledge complexity



Sorenson et al. 2006, Research Policy; Mewes and Broekel 2002, Research Policy

New combinations in science are more impactful if they include conventional knowledge too

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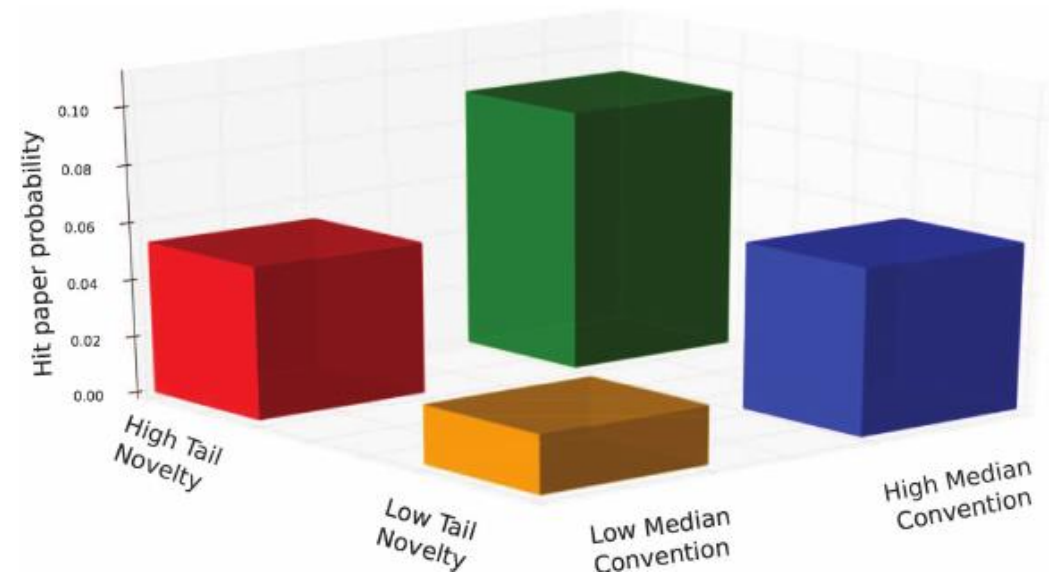
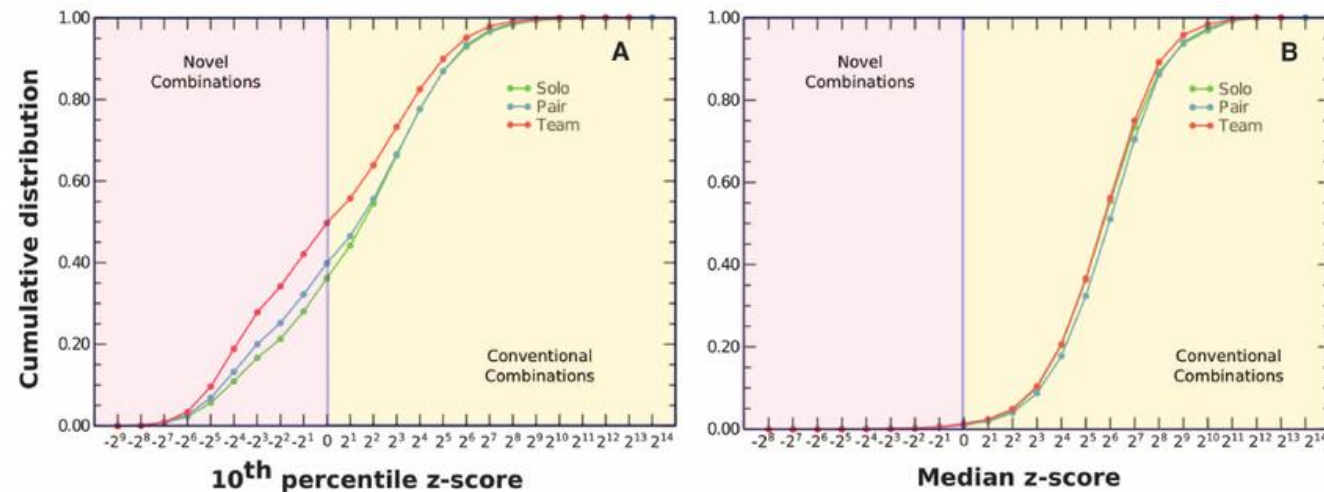
Atypical Combinations and Scientific Impact

Brian Uzzi^{1,2}, Satyam Mukherjee^{1,2}, Michael Stringer^{2,3}, Ben Jones^{1,4,*}

+ See all authors and affiliations

Science 25 Oct 2013:
Vol. 342, Issue 6157, pp. 468-472
DOI: 10.1126/science.1240474

$$Z_{i,j} = \frac{o_{i,j} - e_{i,j}}{\sigma_{i,j}}$$



Negative values represent atypical combination of classes

Research productivity is falling

Are Ideas Getting Harder to Find?

Nicholas Bloom

Charles I. Jones

John Van Reenen

Michael Webb

AMERICAN ECONOMIC REVIEW
VOL. 110, NO. 4, APRIL 2020
(pp. 1104–44)

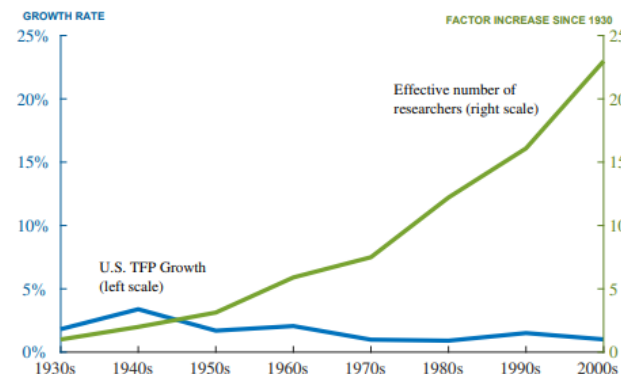
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Article Information

Abstract

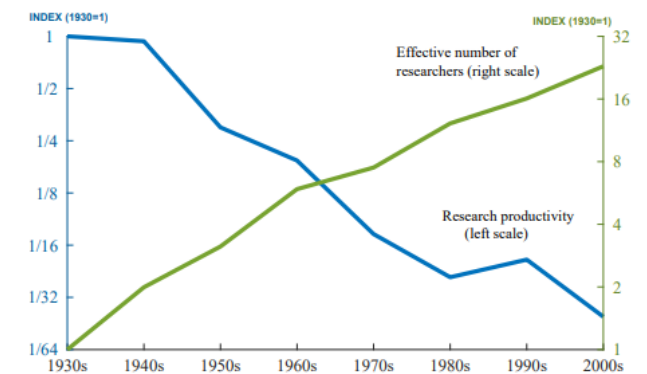
Long-run growth in many models is the product of two terms: the effective number of researchers and their research productivity. We present evidence from various industries, products, and firms showing that research effort is rising substantially while research productivity is declining sharply. A good example is Moore's Law. The number of researchers required today to achieve the famous doubling of computer chip density is more than 18 times larger than the number required in the early 1970s. More generally, everywhere we look we find that ideas, and the exponential growth they imply, are getting harder to find.

Figure 1: Aggregate Data on Growth and Research Effort



Note: The idea output measure is TFP growth, by decade (and for 2000–2014 for the latest observation). For the years since 1950, this measure is the BLS Private Business Sector multifactor productivity growth series, adding back in the contributions from R&D and IPP. For the 1930s and 1940s, we use the measure from Robert Gordon (2016). The idea input measure is gross domestic investment in intellectual property products from the National Income and Product Accounts, deflated by a measure of the nominal wage for high-skilled workers.

Figure 2: Aggregate Evidence on Research Productivity



Note: Research productivity is the ratio of idea output, measured as TFP growth, to research effort. See notes to Figure 1 and the online data appendix. Both research productivity and research effort are normalized to the value of 1 in the 1930s.

AI decreases collaborative knowledge generation

JOURNAL ARTICLE

Large language models reduce public knowledge sharing on online Q&A platforms

R Maria del Rio-Chanona, Nadzeya Laurentsyeva, Johannes Wachs  [Author Notes](#)

PNAS Nexus, Volume 3, Issue 9, September 2024, pgae400,

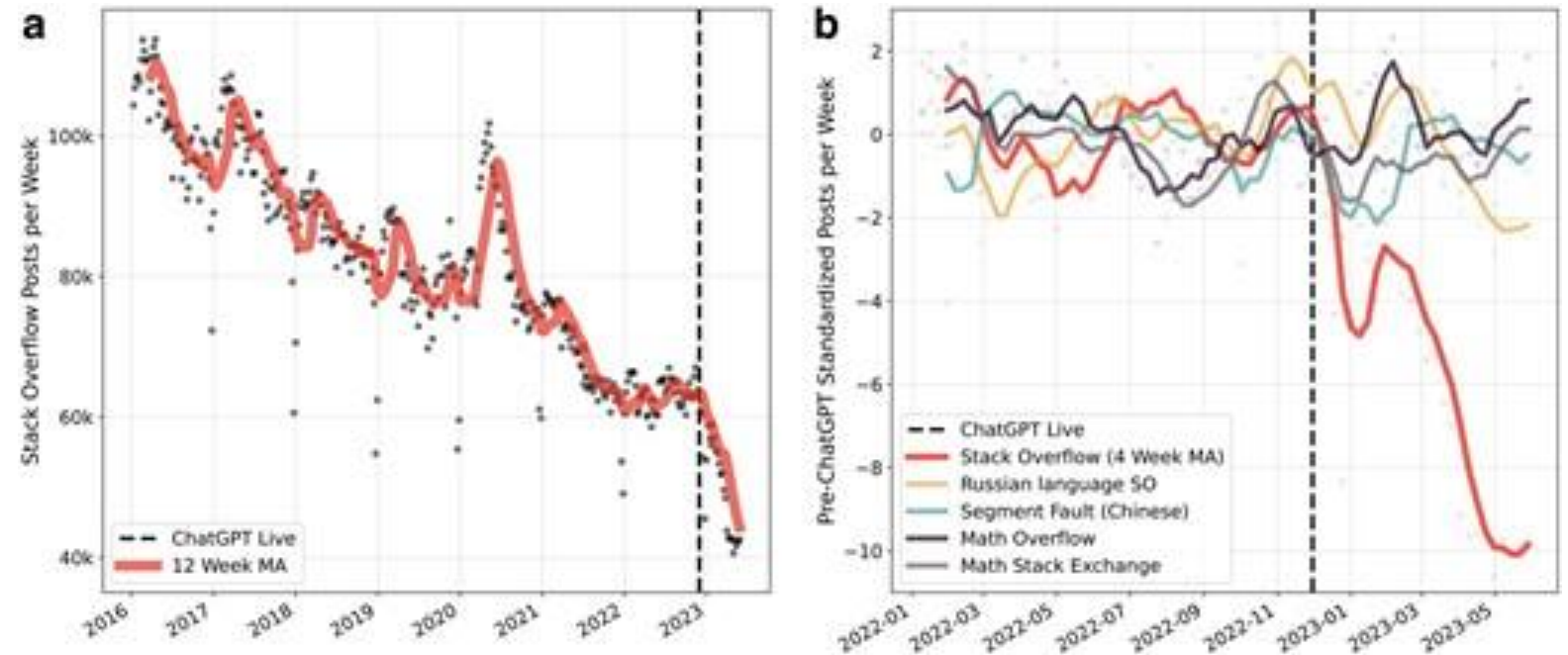
<https://doi.org/10.1093/pnasnexus/pgae400>

Published: 11 September 2024 [Article history](#) ▼

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Abstract

Large language models (LLMs) are a potential substitute for human-generated data and knowledge resources. This substitution, however, can present a significant problem for the training data needed to develop future models if it leads to a reduction of human-generated content. In this work, we document a reduction in activity on Stack Overflow coinciding with the release of ChatGPT, a popular LLM. To test whether this reduction in activity is specific to the introduction of this LLM, we use counterfactuals involving similar human-generated knowledge resources that should not be affected by the introduction of ChatGPT to such extent. Within 6 months of ChatGPT's release, activity on Stack Overflow decreased by 25% relative to its Russian and Chinese counterparts, where access to ChatGPT is limited, and to similar forums for mathematics, where ChatGPT is less capable. We interpret this estimate as a lower bound of the true impact of ChatGPT on Stack Overflow. The decline is larger for posts related to the most widely used programming languages. We find no significant change in post quality, measured by peer feedback, and observe similar decreases in content creation by more and less experienced users alike. Thus, LLMs are not only displacing duplicate, low-quality, or beginner-level content. Our findings suggest that the rapid adoption of LLMs reduces the production of public data needed to train them, with significant consequences.



RESEARCH ARTICLE

International Collaboration and Spatial Dynamics of US Patenting in Central and Eastern Europe 1981-2010

Balázs Lengyel^{1,2,3*}, Mariann Leskó^{3,4}

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Abstract

How did post-socialist transition and a parallel shift in international labor division restructure regional innovation systems in Central and Eastern Europe? This question is increasingly important, because current EU innovation policy is combined with regional development in Smart Specialization Strategies; however, spatial trends of innovation in Central and Eastern Europe are not fully understood which might lead to less than perfectly efficient policy. In this paper we describe the spatial dynamics of inventor activity in the Czech Republic, Hungary, Poland and Slovakia between 1981 and 2010—a period that covers both the late socialist era and the post-socialist transition. Cleaning and analyzing the publicly available data from the United States Patent and Trademark Office we illustrate that Central and Eastern European patents made in international co-operations with partners outside the region receive more citations than those Central and Eastern European patents that lack international co-operation. Furthermore, the technological portfolio of the former patents has become increasingly independent from the technological portfolio of the latter class. A town-level analysis of the applicant-inventor ties reveals that inventors have started to work for foreign assignees in those towns where no innovation activity had been recorded before. However, the positive effect does not last long and patenting seems to be only periodic in the majority of these towns. Therefore, innovation policy in Central and Eastern European countries, as well as in other less developed regions, shall foster synergies between international and domestic collaborations in order to decrease regional disparities in patenting.



OPEN ACCESS

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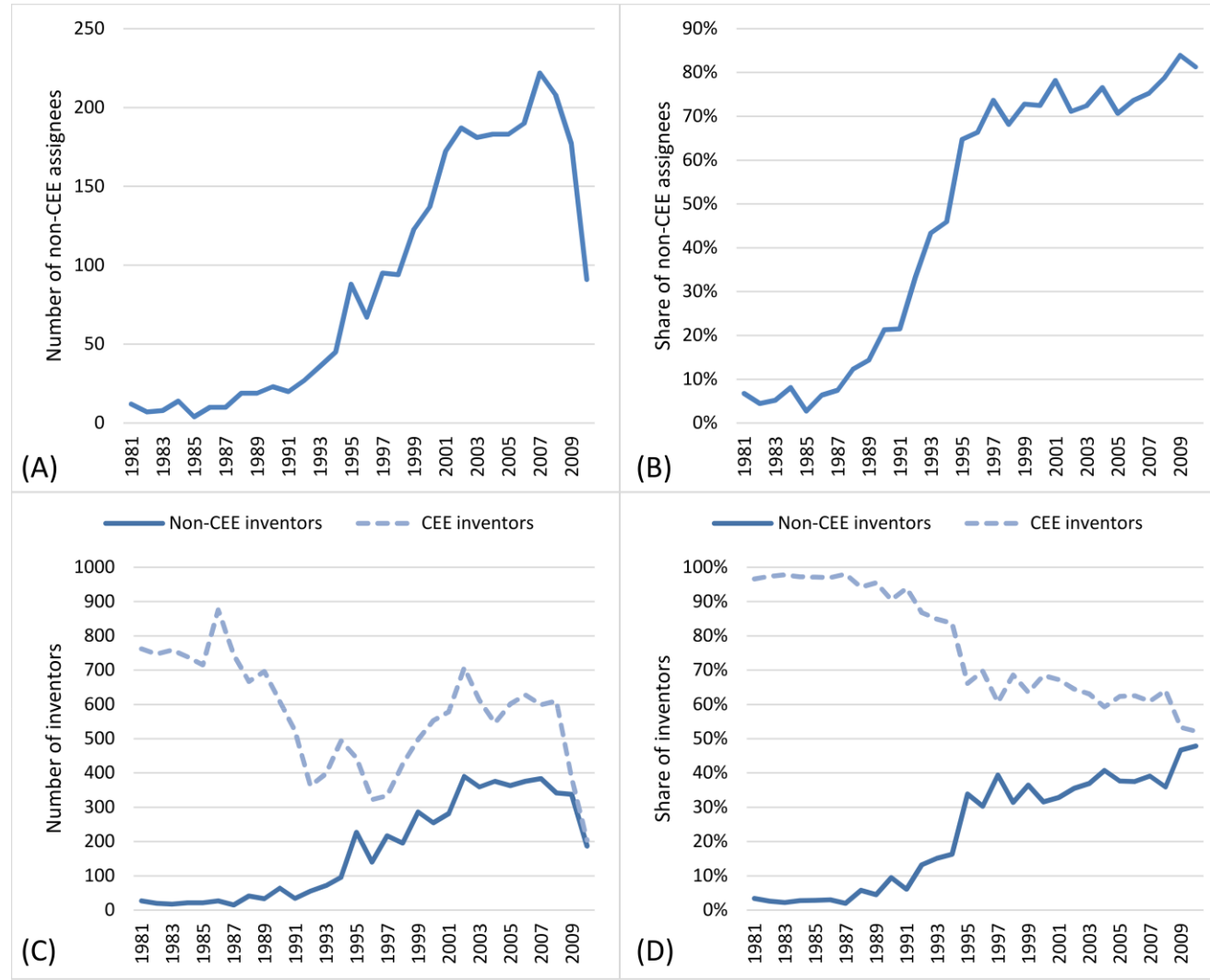
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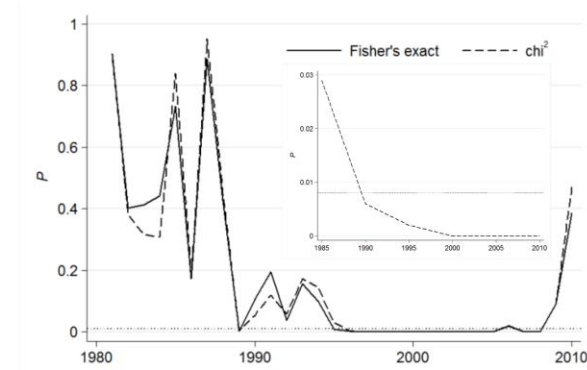
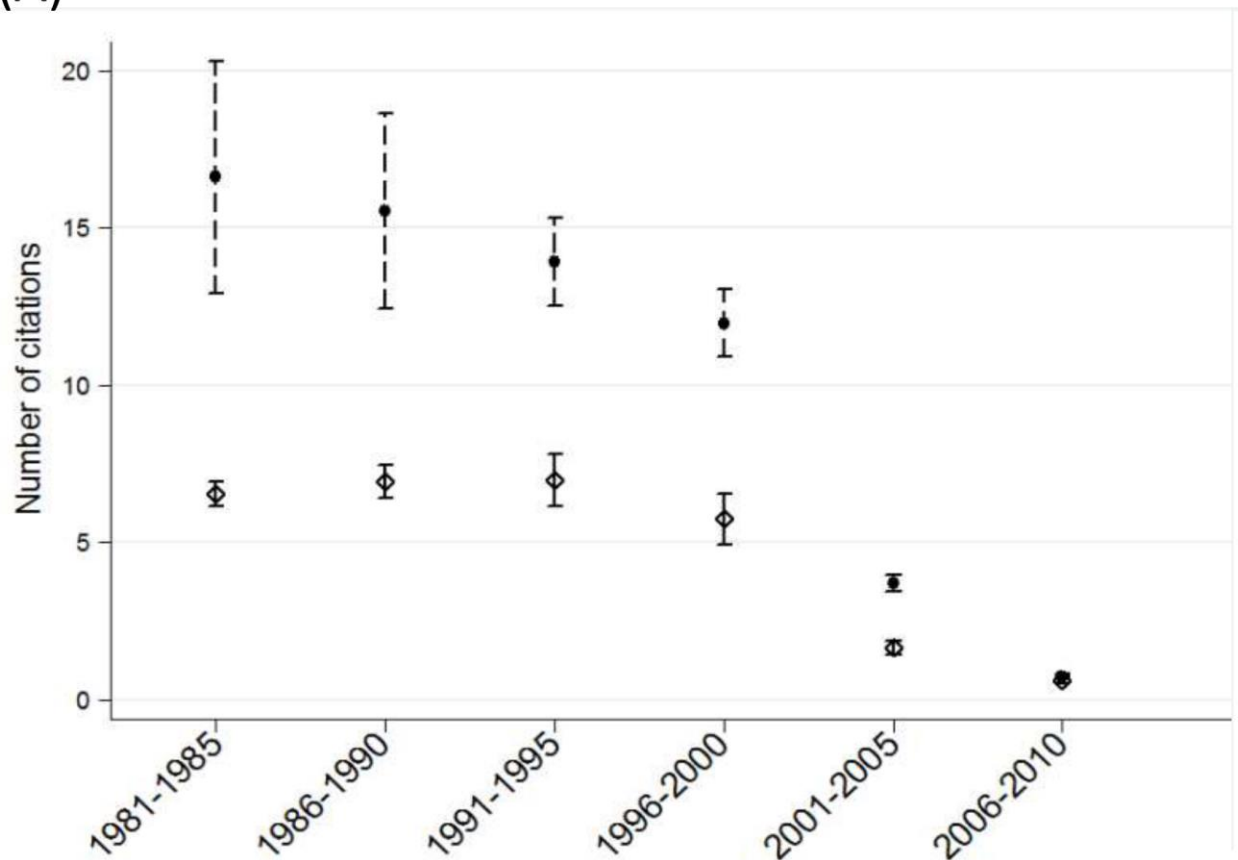
5. Innovation in Central Europe

Foreign firms dominate patenting

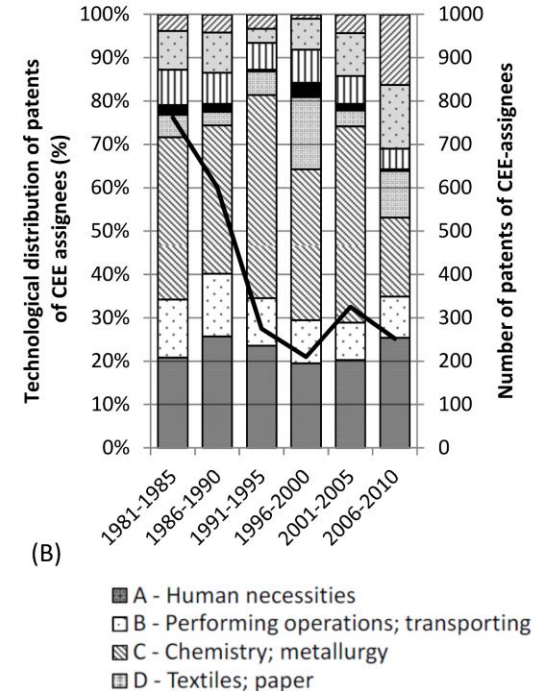


1. Foreign-owned patents are more cited
2. Foreign-ownership changed the technological focus

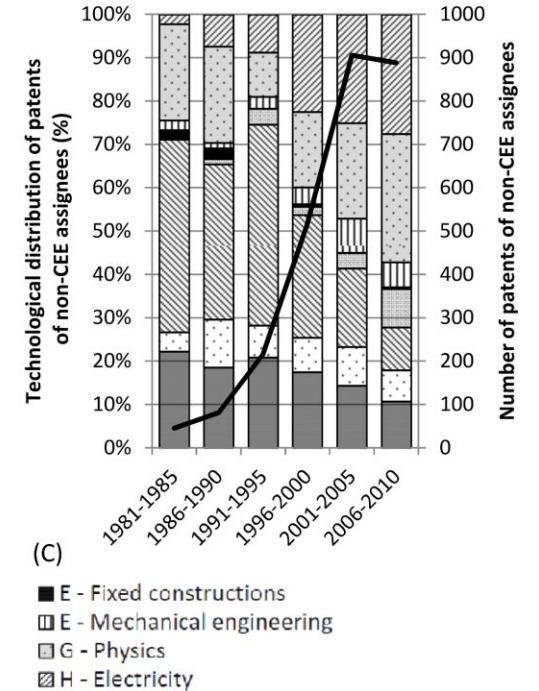
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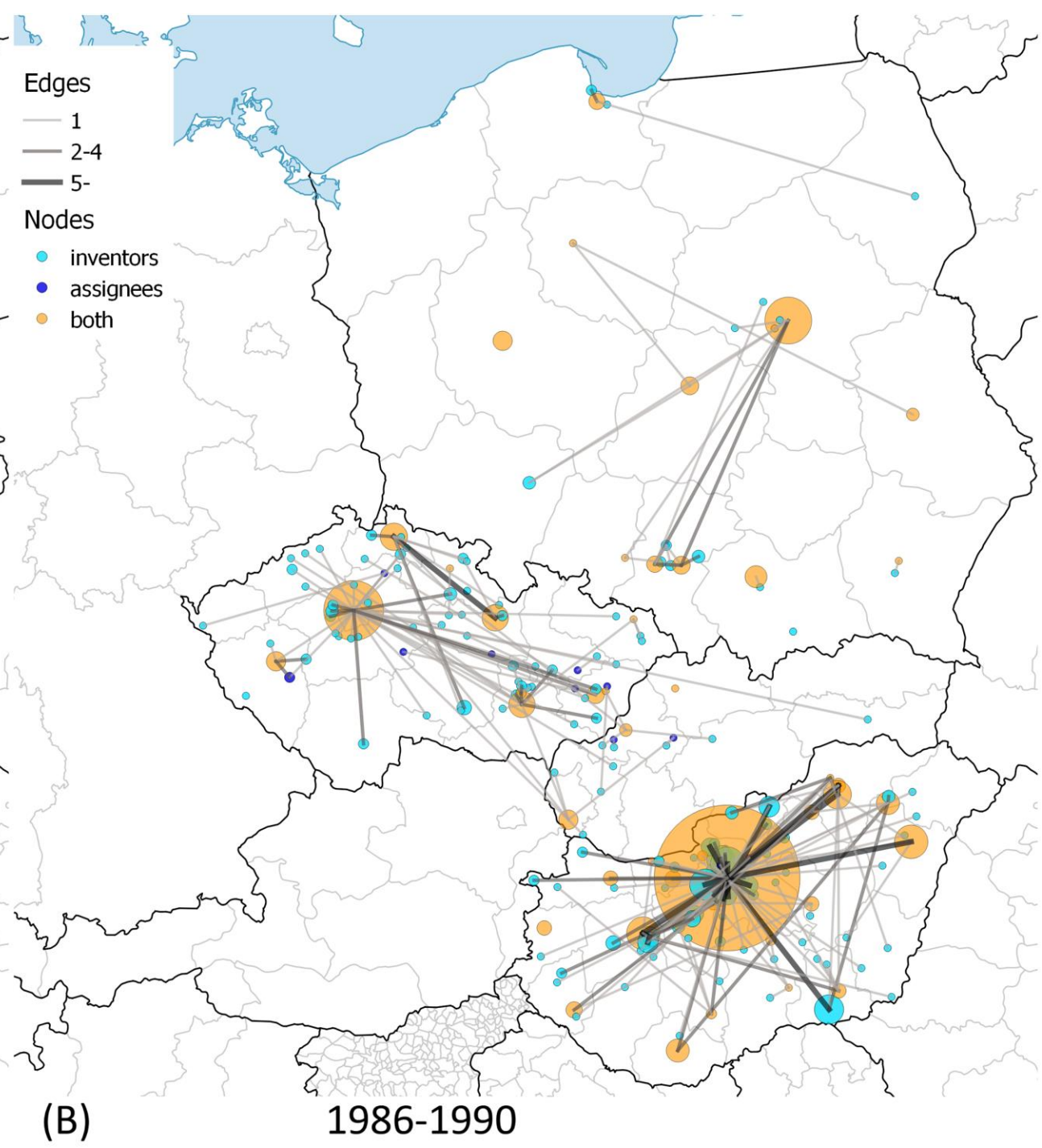
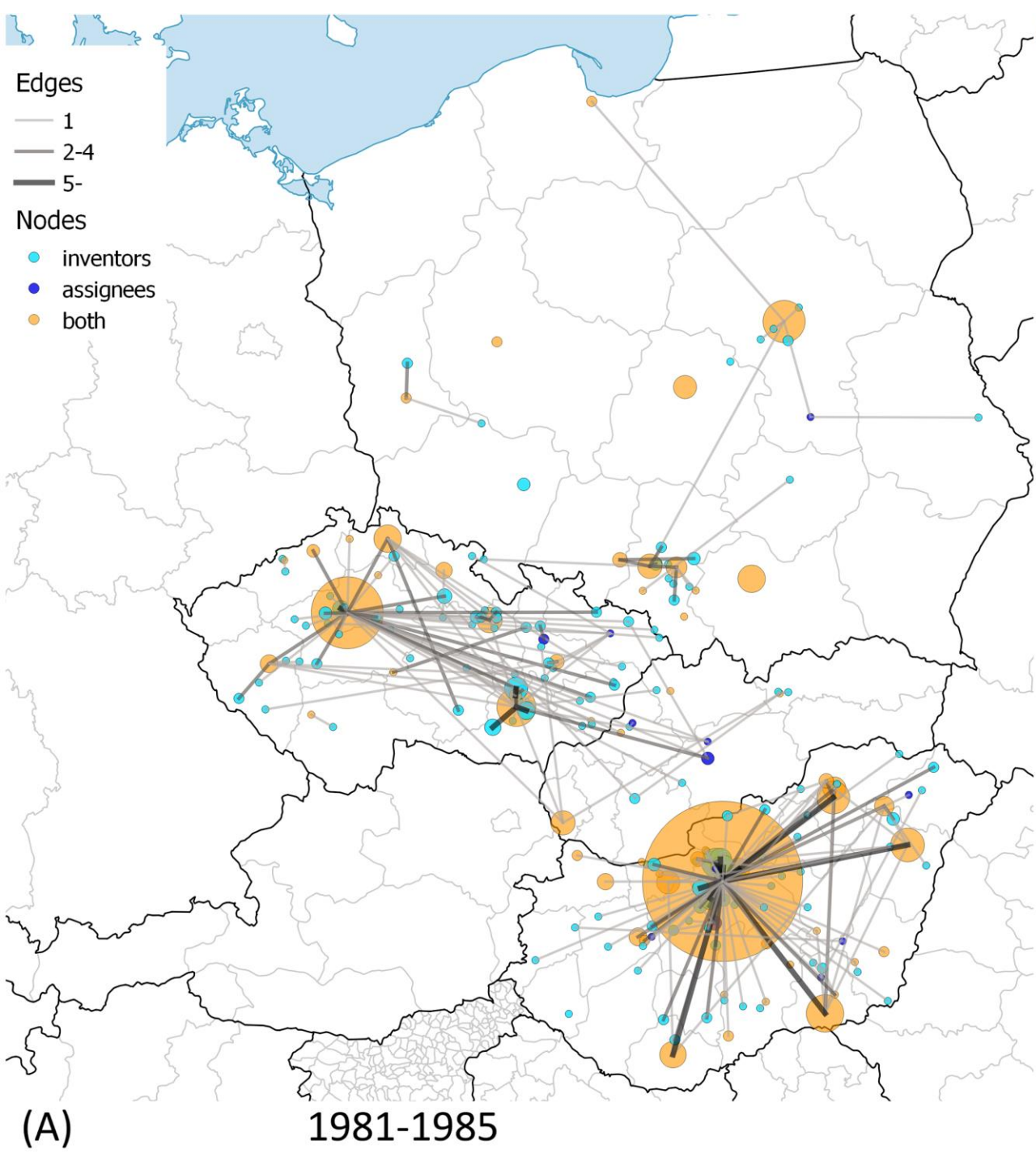
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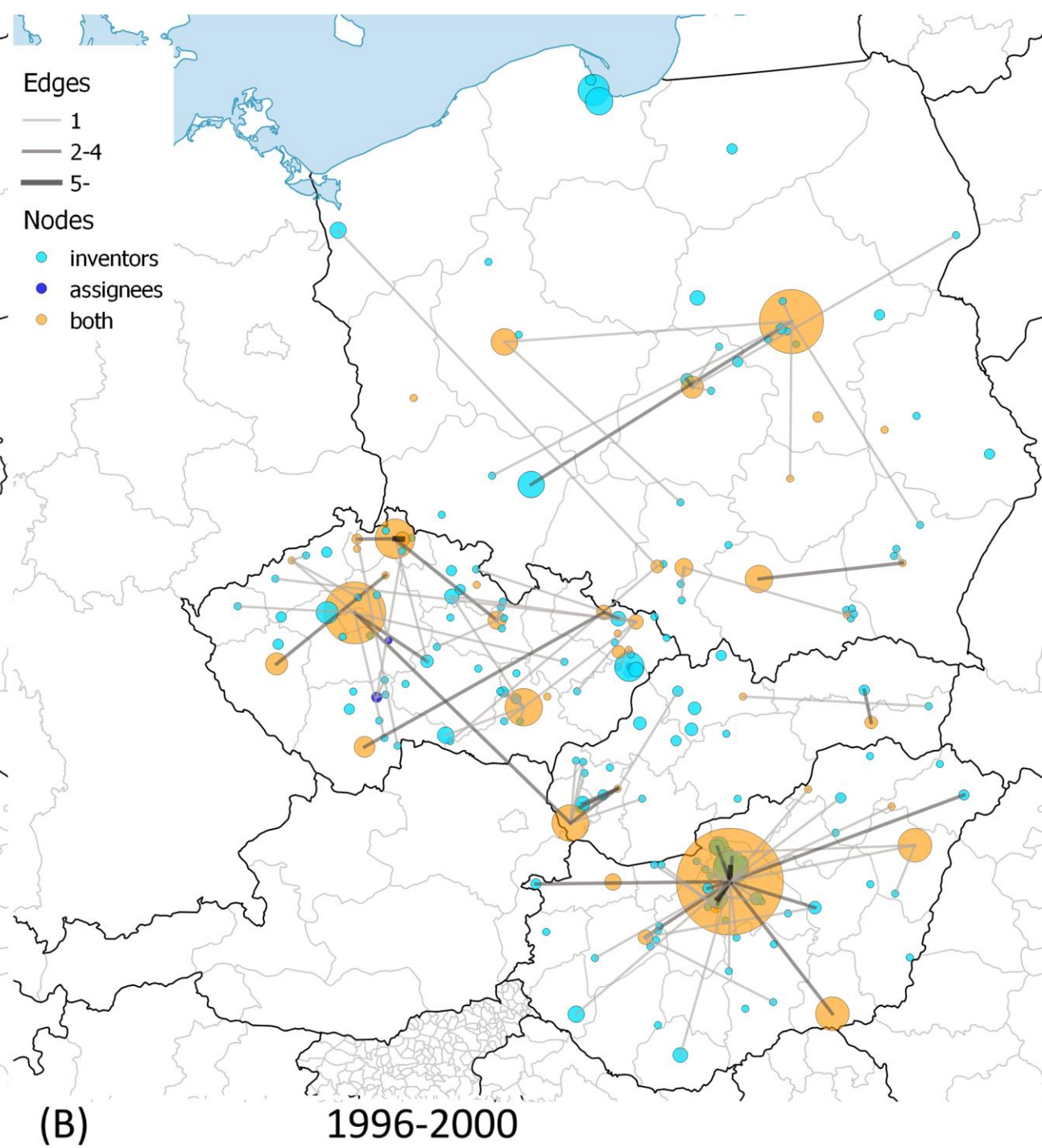
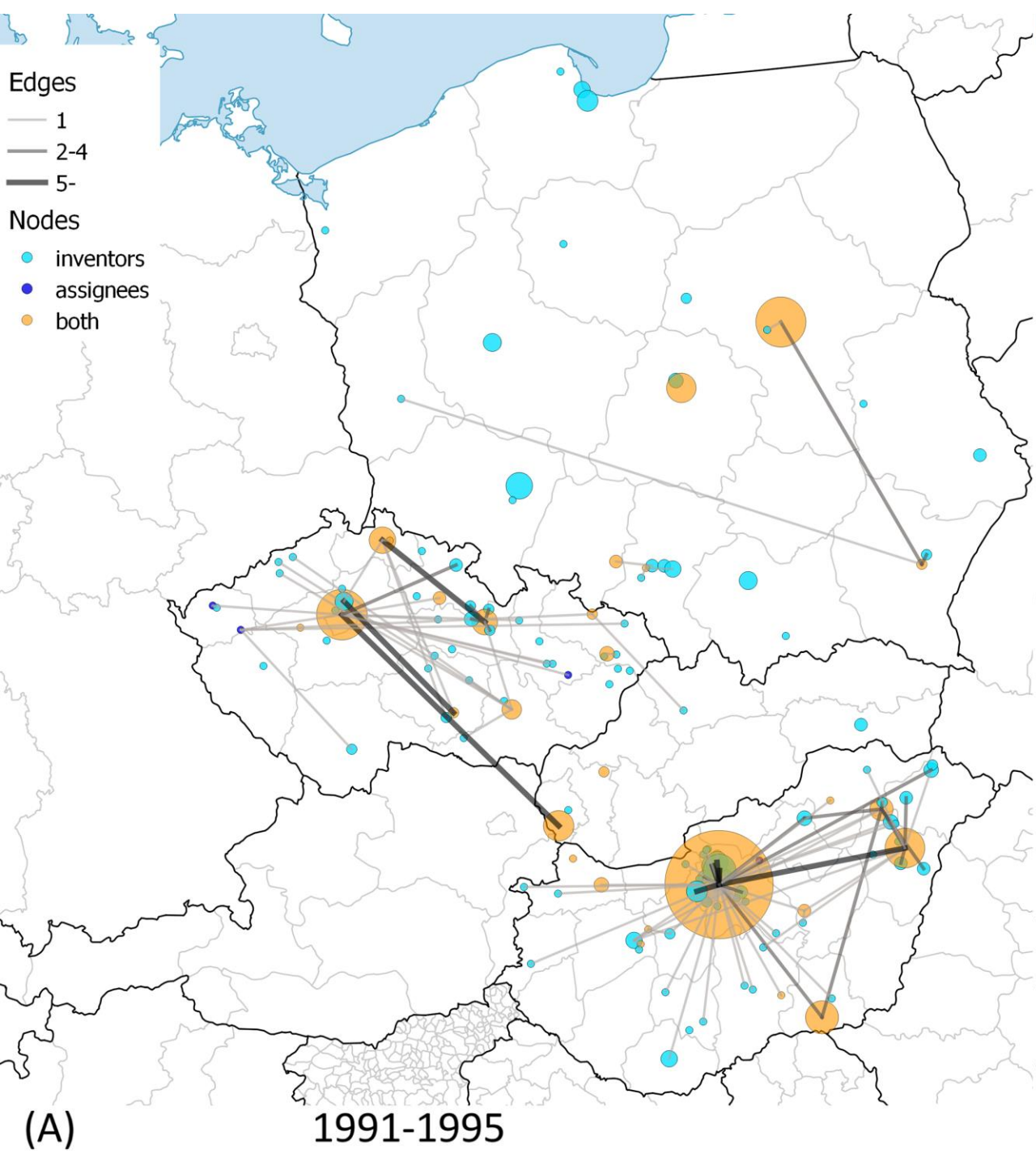


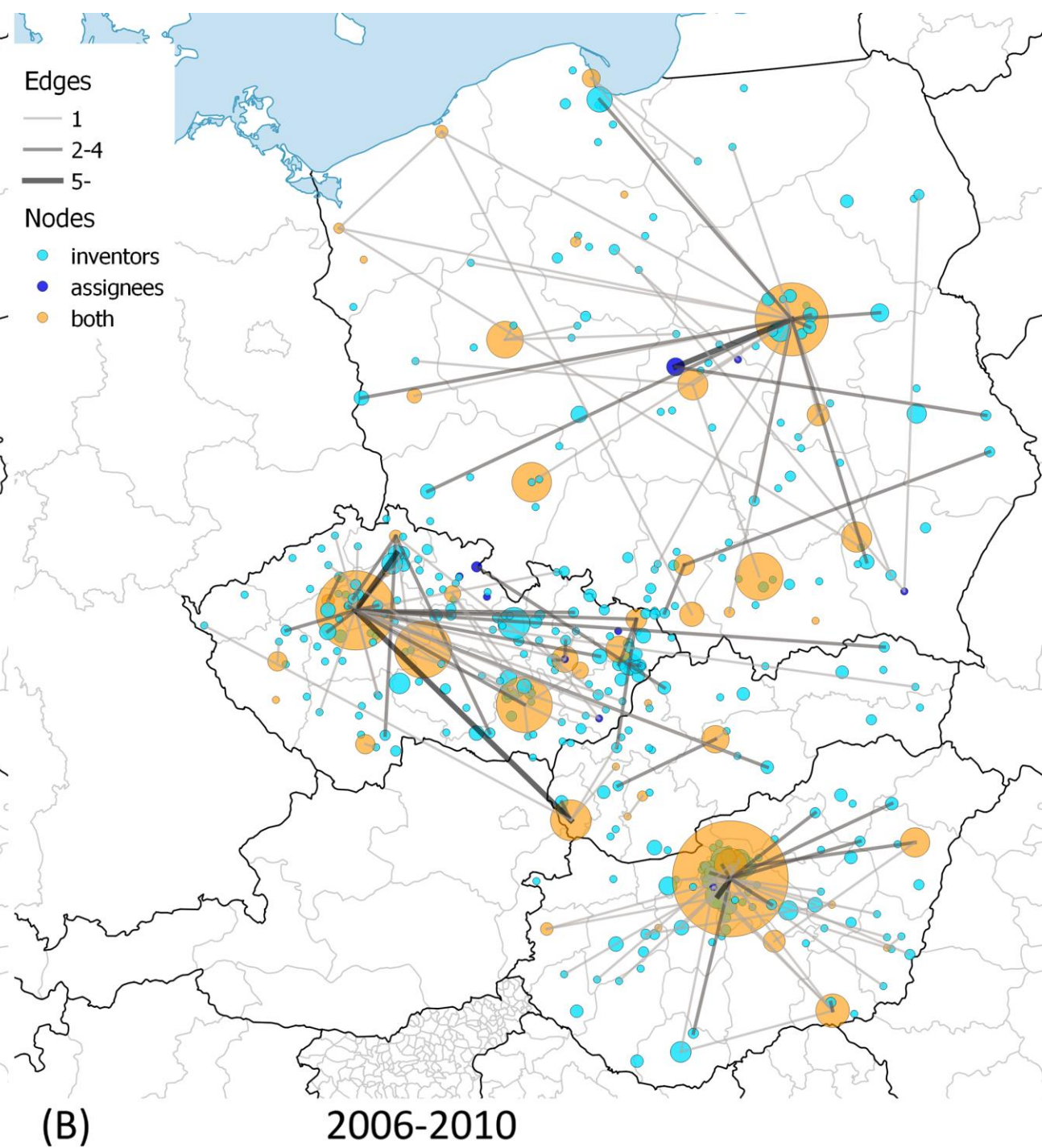
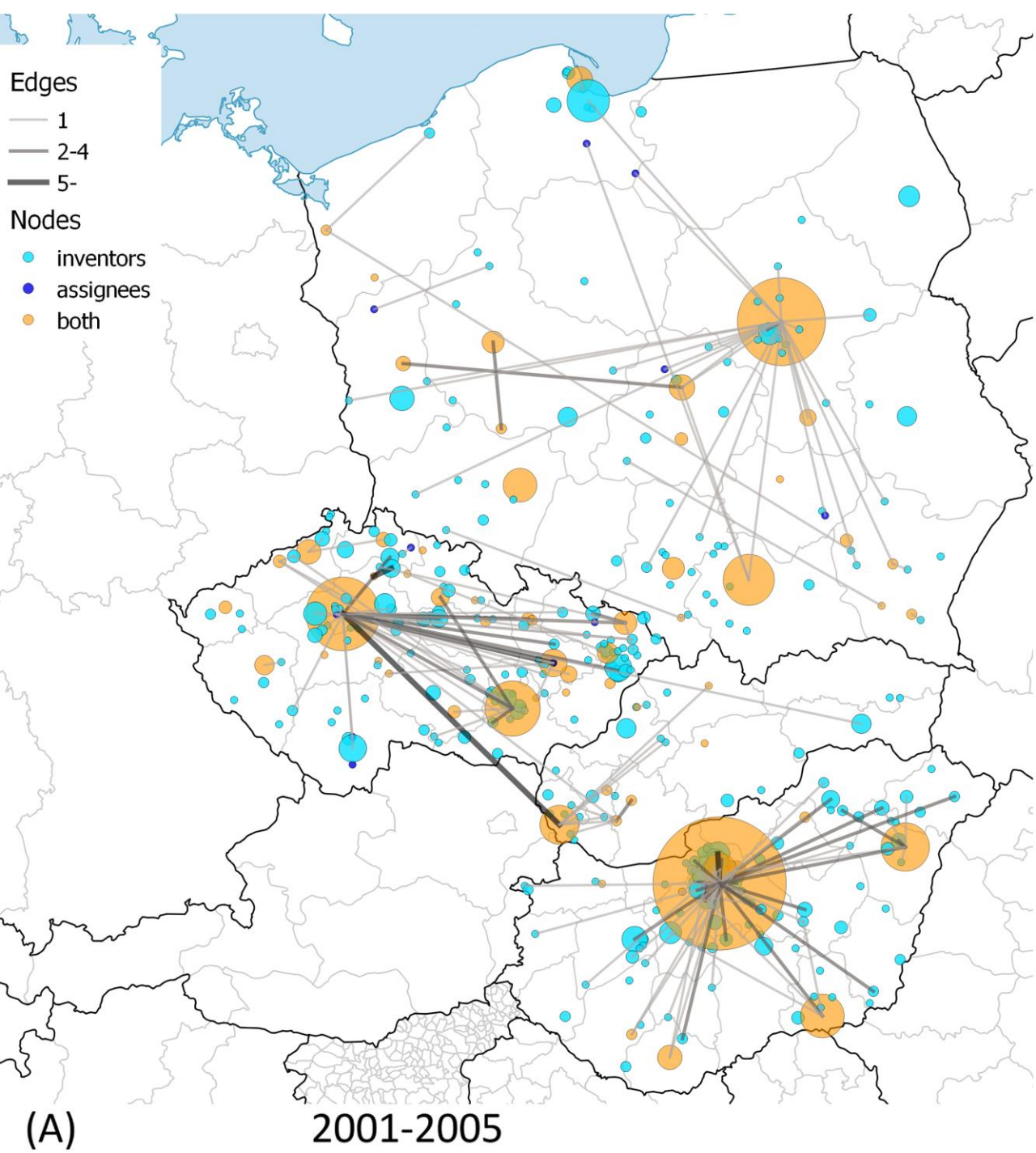
(B)



(C)







Period	1981–1985	1986–1990	1991–1995	1996–2000	2001–2005	2006–2010
Edges	277 (332)	315 (360)	279 (308)	442 (482)	809 (860)	770 (808)
Edges in CEE	242 (297)	249 (294)	111 (140)	121 (161)	195 (246)	172 (210)
Inventor towns in CEE	154	165	112	170	338	367
Assignee towns in CEE	12	11	4	3	11	11
Towns with inventors and assignees in CEE	60	50	29	44	64	64
Non-CEE assignee towns	25	52	98	178	237	199

Note: Town-level self-loops, when the inventor and assignee of the patent are located in the same town, are in parenthesis.

Period	1986–1990	1991–1995	1996–2000	2001–2005	2006–2010
Number of ENTRY	116	67	145	279	236
CEE	95%	61%	53%	36%	37%
NONCEE	3%	37%	42%	54%	60%
CEE and NONCEE	2%	1%	5%	10%	3%

Period	1985–1990	1991–1995	1996–2000	2001–2005	2006–2010
Number of CEE	199	191	80	101	129
INCUMBENT	43.2%	30.4%	37.5%	47.5%	37.2%
EXIT	56.8%	69.6%	62.5%	52.5%	62.8%
Number of NONCEE	5	6	37	74	202
INCUMBENT	60%	16.7%	48.6%	54.1%	36.6%
EXIT	40%	83.3%	51.4%	45.9%	63.4%
Number of CEE and NONCEE	10	18	24	39	71
INCUMBENT	100%	83.3%	87.5%	89.7%	80.3%
EXIT	0%	16.7%	12.5%	10.3%	19.7%

Join the community

- ANETI seminars every Thursday at 3pm
anetilab.org
- Women in Data Science
Beer on Thursday evening (!!!)
- Department of Network Science
Social Lectures Night in November
- Geography of Innovation – join the staff as a student helper!
geoinno2026.org

Jobs at the ANETI Lab

2 research assistant positions

- 10 hours a week (potential increase in the summer)
- 10 months: 2025 November – 2026 August
- Project 1: „Work-from-home in software development”. Data: Slack, Github, Linear records of a mid-size company.
- Project 2: „Social media communication in large cities”. Data: geo-located Twitter in London, Amsterdam, US cities.